Matrix

(1) Definition of Matrix and determinant.

Matrix: Matrix is a set of numbers arranged in rows and columns that forms a rectangular array.

Determinant: The determinant is a scalar value that is a function of entries of a square matrix.

(2) Distinguish between matrix and determinant:

Matrix	Determinant
(1) Matrix can be denoted by [ ], ( ), || ||.	(1) Determinant can be denoted by | |.
(2) Number of rows an columns are not always equal	(2) Numbers of rows and columns are always equal
(3) It has no definite value	(3) It has a definite value
(4) rows and columns can not be interchanged	(4) Rows and columns can be interchanged.

Various types of Matrix :

① ~~Column~~ Row matrix : If in a matrix there is only one row, it is called a row matrix.

Exp : $A = \begin{bmatrix} 1 & 2 & 3 \end{bmatrix}$

② Column matrix : If is a matrix there is only one column, then it is called ~~row~~ column matrix.

Exp : $A = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$

③ Horizontal matrix : If ~~the~~ in a matrix the number of columns is more than the number of rows then it is called the horizontal matrix.

Ex : $\begin{bmatrix} 1 & 2 & 3 & 4 \\ 5 & 6 & 7 & 8 \end{bmatrix} \Longrightarrow 2 \times 4$

(4) **Vertical matrix :** If in a matrix the number of rows is more than the number of columns then it is called the vertical matrix.

Ex: $\begin{bmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{bmatrix} \Rightarrow 3 \times 2$

(5) **Null / Zero Matrix :** If all elements of a matrix are zero then it is called a null or zero matrix.

Ex : $\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} = O$

(6) **Equal Matrix :** Two matrices are equal if they are of the same type, same number of rows and columns and the elements are in the same positions.

Example: $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ $B = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$

$A = B$

(7) **Sub matrix :** A matrix, which is obtained from a given matrix by deleting any number of rows and numbers of columns is called a ~~sub~~ sub matrix of the given matrix.

Ex :

$$A = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 5 & 6 & 7 & 8 \\ 9 & 10 & 11 & 12 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 2 \\ 5 & 6 \end{bmatrix}$$

B is ~~a~~ sub matrix of A.

(8) **Square matrix :** If the number of rows and column equal then the matrix is called a Square matrix.

Ex: $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \Rightarrow 2 \times 2$ (col = row)

(9) **Scalar matrix :** A scalar matrix is a diagonal matrix that has all elements in the diagonal equal to each other.

Exp. $A = \begin{bmatrix} a & 0 & 0 \\ 0 & a & 0 \\ 0 & 0 & a \end{bmatrix}$ → principal Diagonal antry ~~all are~~

(10) **Diagonal Matrix :** A square matrix in which all elements are zero except those in the leading diagonal is known as a diagonal matrix.

Ex : $A = \begin{bmatrix} a & 0 & 0 \\ 0 & b & 0 \\ 0 & 0 & c \end{bmatrix}$ $\quad a \neq b \neq c$

(11) **Unit / Identity matrix :** A square matrix having unity for its elements in the leading diagonal and all other elements as zero is called a unit/identity matrix.

Ex : $A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

(12) **Triangular matrix :**

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 5 & 6 \\ 0 & 0 & 9 \end{bmatrix}$$

$$B = \begin{bmatrix} 1 & 0 & 0 \\ 4 & 5 & 0 \\ 6 & 7 & 8 \end{bmatrix}$$

(13) **Upper Triangular matrix :** Upper Triangular matrix is a Square matrix A whose elements $a_{ij} = 0$ for $~~j~~ . i > j$

Example: $A = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 5 & 6 \\ 0 & 0 & 9 \end{bmatrix}$

(14) **Lower Triangular matrix :** Lower Triangular matrix is a Square matrix A whose elements $a_{ij} = 0$ for $j > i$

Example: $B = \begin{bmatrix} 1 & 0 & 0 \\ 4 & 5 & 0 \\ 7 & 8 & 9 \end{bmatrix}$

(15) **Complex Conjugate Matrix :** The matrix consists of complex numbers is called Complex matrix.

Exap: $A = \begin{bmatrix} 2+i & 0 \\ 2i & -4i \end{bmatrix}$, $\bar{A} = \begin{bmatrix} 2-i & 0 \\ -2i & -4i \end{bmatrix}$

['-' before i term]

(17) **Real Matrix :** A matrix A is called real if it satisfies the relation $\boxed{A = \bar{A}}$.

(18) **Imaginary matrix :** A matrix A is called imaginary if it satisfies the relation, $\boxed{A = -\bar{A}}$.

Ex : $A = \begin{bmatrix} 2i & 4i \\ 7i & -5i \end{bmatrix}$, $\bar{A} = \begin{bmatrix} -2i & -4i \\ -7i & 5i \end{bmatrix}$

$-\bar{A} = \begin{bmatrix} 2i & 4i \\ 7i & -5i \end{bmatrix}$

$\therefore A = -\bar{A}$ satisfy the term

(19) **Transposed of a Matrix :** The transpose of a matrix is created by converting its rows into columns. It is denoted by A^t, A^T, A'.

Example : $A = \begin{bmatrix} 2 & 4 & 6 \\ 8 & 9 & 0 \end{bmatrix}$

$A' = A^t = A^T = \begin{bmatrix} 2 & 8 \\ 4 & 9 \\ 6 & 0 \end{bmatrix}$ row = col

(20) <u>periodic Matrix :</u> $A^{k+1} = A$

(21) <u>Indempotent matrix :</u> $A^2 = A$

(22) <u>Symentic Idempotent matrix :</u> $A = A^t$ and $A = A^2$

(23) <u>Involutory matrix :</u> $A^2 = I$

(24) <u>Nilpotent matrix :</u> $A^m = 0$ & $A^{m-1} \neq 0$

(25) <u>Symmetric matrix :</u> A symmetric matrix is a square matrix that satisfies $\boxed{A^T = A}$

(26) <u>Skew Symmetric matrix :</u> A skew symmetric is a square matrix that satisfies $\boxed{A = -A^T}$

(27) <u>Hermitian Matrix :</u> A square matrix is called hermitian if it satisfies,

$$\boxed{\overline{A^T} = A}$$

(28) <u>Skew Hermitan matrix :</u> A is said to be Skew Hermitan if it satisfies,

$$\overline{A^{T}} = -A$$

(29) <u>Inverse matrix :</u> The inverse of a matrix A.

$$A = A^{-1} = \frac{\text{Adj. of } A}{|A|}$$

(30) <u>Unitary matrix :</u> An unitary matrix is a square complex valued matrix that satisfies the relation

$$A\overline{A}^{T} / A\bar{A} = I$$

(31) <u>Singular matrix :</u> A matrix is singular if only its determinant is ~~not~~ equal 0

$$|A| = \cancel{\neq} 0$$

(32) <u>Non-Singular matrix :</u> A matrix is non-singular if only if determinant is not equal 0

$$|A| \neq 0$$

C.W

Addition of Two Matrix

* $A = \begin{bmatrix} 2 & 4 & 6 \\ 7 & 5 & 3 \end{bmatrix}$ $B = \begin{bmatrix} 4 & 5 & 3 \\ 2 & 1 & 6 \end{bmatrix}$

$\therefore A+B = \begin{bmatrix} 2 & 4 & 6 \\ 7 & 5 & 3 \end{bmatrix} + \begin{bmatrix} 4 & 5 & 3 \\ 2 & 1 & 6 \end{bmatrix}$

$= \begin{bmatrix} 6 & 9 & 9 \\ 9 & 6 & 9 \end{bmatrix}$ Ans

* $3A + 2B = 3\begin{bmatrix} 2 & 4 & 6 \\ 7 & 5 & 3 \end{bmatrix} + 2\begin{bmatrix} 4 & 5 & 3 \\ 2 & 1 & 6 \end{bmatrix}$

$= \begin{bmatrix} 6 & 12 & 18 \\ 21 & 15 & 9 \end{bmatrix} + \begin{bmatrix} 8 & 10 & 6 \\ 4 & 2 & 12 \end{bmatrix}$

$= \begin{bmatrix} 14 & 22 & 24 \\ 25 & 17 & 21 \end{bmatrix}$

Ans

C.W

Matrix Multiplication

condition [number of column of 1st matrix = number of row of 2nd matrix]

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} \times B = \begin{bmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{bmatrix}$$

If $f(x) = x^2 - 5x + 6$ than find $f(A)$,

where,

$$A = \begin{bmatrix} 2 & 0 & 1 \\ 2 & 1 & 6 \\ 1 & -1 & 0 \end{bmatrix}$$

<u>Solve:</u> Given that,

$$A = \begin{bmatrix} 2 & 0 & 1 \\ 2 & 1 & 6 \\ 1 & -1 & 0 \end{bmatrix}$$

also, $f(x) = x^2 - 5x + 6$

$f(A) = A^2 - 5A + 6$

Now, $A^2 = A \times A$

$$= \begin{bmatrix} 2 & 0 & 1 \\ 2 & 1 & 3 \\ 1 & -1 & 0 \end{bmatrix} \times \begin{bmatrix} 2 & 0 & 1 \\ 2 & 1 & 3 \\ 1 & -1 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} 2\times2 + 0\times2 + 1\times1 & 2\times0 + 0\times1 + 1\times(-1) & 2\times1 + 0\times3 + 1\times0 \\ 2\times2 + 1\times2 + 3\times1 & 2\times0 + 1\times1 + 3\times(-1) & 2\times1 + 1\times3 + 3\times0 \\ 1\times2 + (-1)\times2 + 0\times(-1) & 1\times0 + 1\times-1 + 0\times(-1) & 1-3+0 \end{bmatrix}$$

$$= \begin{bmatrix} 4+0+1 & 0+0-1 & 2+0+0 \\ 4+2+3 & 0+1-3 & 2+3+0 \\ 2-2+0 & 0-1-0 & 1-3+0 \end{bmatrix}$$

$$= \begin{bmatrix} 5 & -1 & 2 \\ 9 & -2 & 5 \\ 0 & -1 & -2 \end{bmatrix}$$

$$\therefore F(A) = \begin{bmatrix} 5 & -1 & 2 \\ 9 & -2 & 5 \\ 0 & -1 & -2 \end{bmatrix} - 5\begin{bmatrix} 2 & 0 & 1 \\ 2 & 1 & 3 \\ 1 & -1 & 0 \end{bmatrix} + 6\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 5 & -1 & 2 \\ 9 & -2 & 5 \\ 0 & -1 & -2 \end{bmatrix} - \begin{bmatrix} 10 & 0 & 5 \\ 10 & 5 & 15 \\ 5 & -5 & 0 \end{bmatrix} + \begin{bmatrix} 6 & 0 & 0 \\ 0 & 6 & 0 \\ 0 & 0 & 6 \end{bmatrix}$$

$$= \begin{bmatrix} 11 & -1 & 2 \\ 9 & 4 & 5 \\ 0 & -1 & 4 \end{bmatrix} - \begin{bmatrix} 10 & 0 & 5 \\ 10 & 5 & 15 \\ 5 & -5 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & -1 & -3 \\ -1 & -1 & -10 \\ -5 & 4 & 4 \end{bmatrix}$$ Ans

H.W

Page : 14   Example - 2

If $A = \begin{bmatrix} 3 & 1 & 2 \\ 0 & 1 & 1 \\ 1 & 2 & 0 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 4 \\ 2 & 2 \\ 1 & 0 \end{bmatrix}$

then find AB. Whether BA exists? Give reason

Solv

$$AB = \begin{bmatrix} 3 & 1 & 2 \\ 0 & 1 & 1 \\ 1 & 2 & 0 \end{bmatrix} \times \begin{bmatrix} 1 & 4 \\ 2 & 2 \\ 1 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} 3\times1+1\times2+2\times1 & 3\times4+1\times2+2\times0 \\ 0\times1+1\times2+1\times1 & 0\times4+1\times2+1\times0 \\ 1\times1+2\times2+0\times1 & 1\times4+2\times2+0\times0 \end{bmatrix}$$

$$= \begin{bmatrix} 7 & 14 \\ 3 & 2 \\ 5 & 8 \end{bmatrix}$$

Here A is a matrix of order 3×3 and B is a matrix of order 3×2. Hence BA does not exist as number of columns in B is not equal to the number of rows in A.

Example : 8 (P→ 18)

Find the product of the following two matrices.

$$\begin{bmatrix} 0 & c & -b \\ -c & 0 & a \\ b & -a & 0 \end{bmatrix} \times \begin{bmatrix} a^2 & ab & ac \\ ab & b^2 & bc \\ ac & bc & c^2 \end{bmatrix}$$

Solve.

The required product $= \begin{bmatrix} 0 & c & -b \\ -c & 0 & a \\ b & -a & 0 \end{bmatrix} \times \begin{bmatrix} a^2 & ab & ac \\ ab & b^2 & bc \\ ac & bc & c^2 \end{bmatrix}$

$$= \begin{bmatrix} 0\times a^2 + c\times ab - b.ac & 0\times ab + c.b^2 - b.bc & 0 + c^2b - bc^2 \\ -c.a^2 + 0\times ab + a\times ac & -c\times ab + 0 + a.bc & -c.ac + 0 + a.c^2 \\ b.a^2 - a^2b + 0 & a.b^2 - ab^2 + 0 & b.ac - a.bc + 0 \end{bmatrix}$$

$$= \begin{bmatrix} 0 + abc - abc & 0 + b^2c - b^2c & c^2b - c^2b \\ -a^2c + a^2c & -abc + abc & -ac^2 + ac^2 \\ a^2b - a^2b & ab^2 - ab^2 & abc - abc \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

Ans.

Example : 9 (P – 18)

prove that the product of two matrices

$$\begin{bmatrix} \cos^2\theta & \cos\theta\sin\theta \\ \cos\theta\sin\theta & \sin^2\theta \end{bmatrix} \text{ and } \begin{bmatrix} \cos^2\phi & \cos\phi\sin\phi \\ \cos\phi\sin\phi & \sin^2\phi \end{bmatrix}$$

is zero when θ and ϕ differ by an ~~odd~~ odd multiple of $\frac{1}{2}\pi$.

Solve:

The product =

$$\begin{bmatrix} \cos^2\theta.\cos^2\phi + \cos\theta\sin\theta.\cos\phi\sin\phi & \cos^2\theta.\cos\phi\sin\phi + \cos\theta\sin\theta.\sin^2\phi \\ \cos\theta\sin\theta.\cos^2\phi + \sin^2\theta.\cos\phi\sin\phi & \cos\theta\sin\theta.\cos\phi\sin\phi + \sin^2\theta.\sin^2\phi \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} \cos\theta.\cos\phi(\cos\theta.\cos\phi + \sin\theta.\sin\phi) & \cos\theta\sin\phi(\cos\theta.\cos\phi + \sin\theta.\sin\phi) \\ \sin\theta.\cos\phi(\cos\theta.\cos\phi + \sin\theta.\sin\phi) & \sin\theta\sin\phi(\cos\theta\cos\phi + \sin\theta\sin\phi) \end{bmatrix}$$

$$= \begin{bmatrix} \cos\theta\cos\phi\cos(\theta\sim\phi) & \cos\theta\sin\phi\cos(\theta\sim\phi) \\ \sin\theta\cos\phi(\theta\sim\phi) & \sin\theta\sin\phi\cos(\theta\sim\phi) \end{bmatrix}$$

If $\theta \sim \phi$ = an odd multiple of $\frac{1}{2}\pi$ then $\cos(\theta\sim\phi)=0$ and conseavently the above product is zero.

Example - 11 If A, B, C are three matrices such that,

$$A = [x, y, z],\ B = \begin{bmatrix} a & h & g \\ h & b & f \\ g & f & c \end{bmatrix},\ C = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

evaluate ABC.

Solve

$$AB = [x, y, z] \times \begin{bmatrix} a & h & g \\ h & b & f \\ g & f & c \end{bmatrix}$$

$$= [ax + hy + zg \quad hx + by + fz \quad gx + fy + cz]$$

$$ABC = [ax + hy + zg \quad hx + by + fz \quad gx + fy + cz] \times \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

$$= [x(ax + hy + gz) + y(hx + by + fz) + z(gx + fy + cz)]$$

$$= \cancel{[ax^2 + hy^2 + gz^2}$$

$$= [ax^2 + by^2 + cz^2 + 2hxy + 2gzx + 2fyz]$$

Ans.

Elementary row operation metod

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Example - 1 : Find the inverse of the matrix $A = \begin{bmatrix} 1 & -3 & 2 \\ 2 & 0 & 0 \\ 1 & 4 & 1 \end{bmatrix}$

Solve Given that,

$$A = \begin{bmatrix} 1 & -3 & 2 \\ 2 & 0 & 0 \\ 1 & 4 & 1 \end{bmatrix}$$

Now, $A | I$

$$\sim \begin{bmatrix} 1 & -3 & 2 \\ 2 & 0 & 0 \\ 1 & 4 & 1 \end{bmatrix} \bigg| \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\sim \begin{bmatrix} 2 & 0 & 0 \\ 1 & -3 & 2 \\ 1 & 4 & 1 \end{bmatrix} \bigg| \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad \left[\text{Applying } R_1 \rightleftharpoons R_2\right]$$

$$\sim \begin{bmatrix} 1 & 0 & 0 \\ 1 & -3 & 2 \\ 1 & 4 & 1 \end{bmatrix} \bigg| \begin{bmatrix} 0 & \frac{1}{2} & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad \left[\text{Applying } \frac{R_1}{2} = R_1'\right]$$

$$\sim \left[\begin{array}{ccc} 1 & 0 & 0 \\ 0 & -3 & 2 \\ 0 & 4 & 1 \end{array}\right] \Bigg| \left[\begin{array}{ccc} 0 & \frac{1}{2} & 0 \\ 1 & -\frac{1}{2} & 0 \\ 0 & -\frac{1}{2} & 1 \end{array}\right]$$

[Applying $R_2 - R_1 = R_2'$, $R_3 - R_1 = R_3'$]

$$\sim \left[\begin{array}{ccc} 1 & 0 & 0 \\ 0 & -11 & 0 \\ 0 & 4 & 1 \end{array}\right] \Bigg| \left[\begin{array}{ccc} 0 & \frac{1}{2} & 0 \\ 1 & \frac{1}{2} & -2 \\ 0 & -\frac{1}{2} & 1 \end{array}\right]$$

[Apply $R_2 - 2R_3 = R_2'$]

$-2 - 1$

$$\sim \left[\begin{array}{ccc} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 4 & 1 \end{array}\right] \Bigg| \left[\begin{array}{ccc} 0 & \frac{1}{2} & 0 \\ \frac{1}{-11} & -\frac{1}{22} & \frac{2}{11} \\ 0 & -\frac{1}{2} & 1 \end{array}\right]$$

[Apply $\frac{R_2}{-11} = R_2'$]

$$\sim \left[\begin{array}{ccc} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{array}\right] \Bigg| \left[\begin{array}{ccc} 0 & \frac{1}{2} & 0 \\ -\frac{1}{11} & -\frac{1}{22} & \frac{2}{11} \\ \frac{4}{11} & -\frac{7}{22} & \frac{3}{11} \end{array}\right]$$

[$R_3 - 4R_2 = R_3'$]

$1 - \frac{8}{11} = \frac{11-8}{11} = \frac{3}{11}$

$\frac{2\times4}{11}$

$1 - \frac{8}{11}$

$\frac{11-8}{11}$

$\sim I \mid A^{-1}$

Hence, $A^{-1} = \left[\begin{array}{ccc} 0 & \frac{1}{2} & 0 \\ -\frac{1}{11} & -\frac{1}{22} & \frac{2}{11} \\ \frac{4}{11} & -\frac{7}{22} & \frac{3}{11} \end{array}\right]$

Ans

H.W

Example - 2 $\quad A = \begin{bmatrix} 1 & 2 & 1 \\ 3 & 2 & 3 \\ 1 & 1 & 2 \end{bmatrix}$ evaluate A^{-1}

Solve: ~~A+I~~ Here given,

$$A = \begin{bmatrix} 1 & 2 & 1 \\ 3 & 2 & 3 \\ 1 & 1 & 2 \end{bmatrix}$$

Now, $A | I$

$$\begin{bmatrix} 1 & 2 & 1 \\ 3 & 2 & 3 \\ 1 & 1 & 2 \end{bmatrix} \Bigg| \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\sim \begin{bmatrix} 1 & 2 & 1 \\ 0 & -4 & 0 \\ 0 & -1 & 1 \end{bmatrix} \Bigg| \begin{bmatrix} 1 & 0 & 0 \\ -3 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix} \quad \left[\text{Applying } R_2 - 3R_1 = R_2' \text{ and } R_3 - R_1 = R_3'\right]$$

$$\sim \begin{bmatrix} 1 & 3 & 0 \\ 0 & 1 & 0 \\ 0 & -1 & 1 \end{bmatrix} \Bigg| \begin{bmatrix} 2 & 0 & -1 \\ 3/4 & -1/4 & 0 \\ -1 & 0 & 1 \end{bmatrix}$$

$\left[\text{Applying } R_1 - R_3 = R_1', \ R_2/_{-4} = R_2'\right]$

$$\sim \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \Bigg| \begin{bmatrix} -\frac{1}{4} & \frac{3}{4} & -1 \\ \frac{3}{4} & -\frac{1}{4} & 0 \\ -\frac{1}{4} & -\frac{1}{4} & 1 \end{bmatrix}$$

[Apply $R_1 - R_2 = R_1'$ & $R_3 + R_2 = R_3'$]

$$\therefore A^{-1} = \begin{bmatrix} -\frac{1}{4} & \frac{3}{4} & -1 \\ \frac{2}{4} & -\frac{1}{4} & 0 \\ -\frac{1}{4} & -\frac{1}{4} & 1 \end{bmatrix}$$

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Canonical form / normal form / orthogonal of a matrix :

Definition : Every non-zero $m \times n$ matrix A can be reduced by means of elements transformations to the form $\begin{bmatrix} I_r & 0 \\ 0 & 0 \end{bmatrix}$,

where I_r is the $r \times r$ identity matrix and the remaining sub-matrix are zero matrix. The ~~avobe~~ above form is called the canonical form or orthogonal from of matrix A.

Rank of Matrix:

The maximum number of linearly independent columns or rows of a matrix is called the rank of a matrix.

Example:

$$\begin{bmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Rank of matrix: If $m \times n$ matrix A is reduced to the canonical form or normal form $\begin{bmatrix} I_r & 0 \\ 0 & 0 \end{bmatrix}$ by the application of elementary row or column operations, then r, the order of the identity sub-matrix I_r is to rank of the matrix A.

Example - 1

Reduce $A = \begin{bmatrix} 1 & 2 & -1 & 4 \\ 2 & 4 & 3 & 5 \\ 1 & 2 & 3 & 4 \\ -1 & -2 & 6 & -7 \end{bmatrix}$ to the canonical form.

Solve

Given that,

$$A = \begin{bmatrix} 1 & 2 & -1 & 4 \\ 2 & 4 & 3 & 5 \\ 1 & 2 & 3 & 4 \\ -1 & -2 & 6 & -7 \end{bmatrix}$$

Applying

$$\begin{bmatrix} r_2 - 2r_1 = r_2' \\ r_3 - r_1 = r_3' \\ r_4 + r_1 = r_4' \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 2 & -1 & 4 \\ 0 & 0 & 5 & -3 \\ 0 & 0 & 4 & 0 \\ 0 & 0 & 5 & -3 \end{bmatrix}$$

Apply

$$\begin{bmatrix} c_2 - 2c_1 = c_2' \\ c_3 + c_1 = c_3' \\ c_4 - 4c_1 = c_4' \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 5 & -3 \\ 0 & 0 & 4 & 0 \\ 0 & 0 & 5 & -3 \end{bmatrix}$$

Apply

$$\begin{bmatrix} r_3' = \frac{r_3}{4} \\ r_4' = r_4 - r_2 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 5 & -3 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & -3 & 5 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \quad [C_2 \rightleftharpoons C_4]$$

$$\approx \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & -3 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \quad \left[\text{Applying } R_2' = R_2 - 5R_3\right]$$

$$\approx \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \quad \left[\text{Applying, } R_2' = \frac{R_2}{-3}\right]$$

$$\approx \begin{bmatrix} I_3 & 0 \\ 0 & 0 \end{bmatrix}$$ which is the required canonical form

C.W

Example : 9(a) Reduce the matrix A to the normal form and hence find the rank of the matrix A, where,

$$A = \begin{bmatrix} 0 & 1 & -3 & -1 \\ 1 & 0 & 1 & 1 \\ 3 & 1 & 0 & 2 \\ 1 & 1 & -2 & 0 \end{bmatrix}$$

Solve

Given that,

$$A = \begin{bmatrix} 0 & 1 & -3 & -1 \\ 1 & 0 & 1 & 1 \\ 3 & 1 & 0 & 2 \\ 1 & 1 & -2 & 0 \end{bmatrix}$$

Applying $C_3' = C_3 - C_1$ & $C_4' = C_4 - C_1$

$$A = \begin{bmatrix} 0 & 1 & -3 & -1 \\ 1 & 0 & 0 & 0 \\ 3 & 1 & -3 & -1 \\ 1 & 1 & -3 & -1 \end{bmatrix}$$

Applying $r_4' = r_4 - r_1$ & $r_3' = r_3 - r_1$

$$= \begin{bmatrix} 0 & 1 & -3 & -1 \\ 1 & 0 & 0 & 0 \\ 3 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix}$$

Applying $C_3' = C_3 + 3C_2$ & $C_4' = C_2 + C_4$

$$= \begin{bmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 3 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix}$$

Apply $R_3' = R_3 - 3R_4$,

$$= \begin{bmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix}$$

Applying $R_4 - R_2 = R_4'$

$$= \begin{bmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \longrightarrow$$ Find Rank বললে এই পর্যন্ত

$$= \begin{bmatrix} I_2 & 0 \\ 0 & 0 \end{bmatrix}$$ [Applying $R_1 \rightleftarrows R_2$]

which is our required canonical form also hence number of non-zero rows is 2. So, the rank of the give matrix is 2

H.W

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Example – 3(b)

Find the rank of the matrix $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 4 \\ 3 & 5 & 7 \end{bmatrix}$ after reducing it.

Solve Here given, $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 4 \\ 3 & 5 & 7 \end{bmatrix}$

$$= \begin{bmatrix} 1 & 1 & 1 \\ 2 & 1 & 1 \\ 3 & 2 & 2 \end{bmatrix} \qquad \text{Applying} \begin{bmatrix} C_2' = C_2 - C_1 \\ C_3' = C_3 - C_2 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 1 & 1 \\ 1 & 0 & 0 \\ 1 & 1 & 1 \end{bmatrix} \qquad \text{Applying} \begin{bmatrix} R_2' = R_2 - R_1 \\ R_3' = R_3 - R_2 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 1 & 1 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \qquad \begin{bmatrix} \text{Applying} \\ R_3' = R_3 - R_1 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \quad \left[\begin{array}{l} \text{Applying} \\ C_1' = C_1 - C_2 \\ C_3' = C_3 - C_2 \end{array}\right]$$

$$= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \quad [R_1 \rightleftharpoons R_2]$$

$$= \begin{bmatrix} I_2 & 0 \\ 0 & 0 \end{bmatrix}$$

Hence the Rank of A is 2.

Example - 7 (Page - 23)

By elementary operations, Find the Rank of the matrix.

$$A = \begin{bmatrix} 2 & 3 & -1 & -1 \\ 1 & -1 & -2 & -4 \\ 3 & 1 & 3 & -2 \\ 6 & 3 & 0 & -7 \end{bmatrix}$$

Here given,

$$A = \begin{bmatrix} 2 & 3 & -1 & -1 \\ 1 & -1 & -2 & -4 \\ 3 & 1 & 3 & -2 \\ 6 & 3 & 0 & -7 \end{bmatrix}$$

$$= \begin{bmatrix} 6 & 3 & 0 & -7 \\ 1 & -1 & -2 & -4 \\ 3 & 1 & 3 & -2 \\ 6 & 3 & 0 & -7 \end{bmatrix} \quad \begin{bmatrix} \text{Apply} \\ R_1' = R_1 + R_2 + R_3 \end{bmatrix}$$

$$= \begin{bmatrix} 6 & 3 & 0 & -7 \\ 1 & -1 & -2 & -4 \\ 3 & 1 & 3 & -2 \\ 0 & 0 & 0 & 0 \end{bmatrix} \quad \begin{bmatrix} \text{Applying} \\ R_4' = R_4 - R_1 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 9 & 12 & 17 \\ 1 & -1 & -2 & -4 \\ 0 & 4 & 9 & 10 \\ 0 & 0 & 0 & 0 \end{bmatrix} \quad \begin{bmatrix} \text{Applying} \\ R_1' = R_1 - 6R_2 \\ R_3' = R_3 - 3R_2 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 9 & 12 & 17 \\ 1 & 0 & 0 & 0 \\ 0 & 4 & 9 & 10 \\ 0 & 0 & 0 & 0 \end{bmatrix} \quad \begin{bmatrix} \text{Applying} \\ C_2' = C_2 + C_1 \\ C_3' = C_3 + 3C_1 \\ C_4' = C_4 + 4C_1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 9 & 12 & 17 \\ 0 & 4 & 9 & 10 \\ 0 & 0 & 0 & 0 \end{bmatrix} \quad \begin{bmatrix} R_1 \rightleftharpoons R_2 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & -6 & -3 \\ 0 & 4 & 9 & 10 \\ 0 & 0 & 0 & 0 \end{bmatrix} \quad \left[\text{Applying } r_2' = r_2 - 2r_3\right]$$

$$= \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 4 & 33 & 22 \\ 0 & 0 & 0 & 0 \end{bmatrix} \quad \left[\begin{array}{l}\text{Apply } c_3' = c_3 + 6c_2 \\ \quad c_4' = c_4 + 3c_2\end{array}\right]$$

$$= \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 4 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} \quad \left[\begin{array}{l}\text{Apply} \\ c_2' = c_2 - 4c_3 \\ c_4 = c_4 - c_3\end{array}\right]$$

$$= \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} I_3 & 0 \\ 0 & 0 \end{bmatrix}$$

Hence the rank of the given matrix = 3

Example - 11(a)

Find the rank of matrix $A = \begin{bmatrix} 3 & -2 & 0 & -1 \\ 0 & 2 & 2 & 1 \\ 1 & -2 & -3 & 2 \\ 0 & 1 & 2 & 1 \end{bmatrix}$

Here given,

$$A = \begin{bmatrix} 3 & -2 & 0 & -1 \\ 0 & 2 & 2 & 1 \\ 1 & -2 & -3 & 2 \\ 0 & 1 & 2 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & -2 & -3 & 2 \\ 0 & 2 & 2 & 1 \\ 3 & -2 & 0 & -1 \\ 0 & - & & \end{bmatrix}$$

[$r_1 \rightleftharpoons r_3$ And Apply $r_4' = r_4 - r_2$]

$$= \begin{bmatrix} 0 & 4 & 9 & -7 \\ 0 & 1 & 0 & 0 \\ 1 & -2 & -3 & 2 \\ 0 & 1 & 2 & 1 \end{bmatrix}$$

[Applying $r_1' = r_1 - 3r_3$, $r_2' = r_2 - r_4$]

$$= \begin{bmatrix} 1 & -2 & -3 & 2 \\ 0 & 1 & 0 & 0 \\ 0 & 4 & 9 & -7 \\ 0 & 0 & 2 & 1 \end{bmatrix}$$

[$r_1 \rightleftharpoons r_3$ and Apply $r_4' = r_4 - r_2$]

$$\Rightarrow \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 4 & 9 & -7 \\ 0 & 0 & 2 & 1 \end{bmatrix}$$ Apply $C_2' = C_2 + 2C_1$, $C_3' = C_3 + 3C_1$, $C_4' = C_4 - 2C_1$

$$= \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 9 & -7 \\ 0 & 0 & 2 & 1 \end{bmatrix}$$ Apply $R_3' = R_3 - 4R_2$

$$\Rightarrow \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 23 & 0 \\ 0 & 0 & 2 & 1 \end{bmatrix}$$ Apply, $R_3' = R_3 + 7R_4$

$$= \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 2 & 1 \end{bmatrix}$$ Apply $R_3' = \frac{R_3}{23}$

$$\Rightarrow \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$ $C_3'' = C_3 - 2C_4$

$= [I_4]$ Hence the rank of A is 4.

Example-14 -(page-31)

Find the rank of the matrix $A = \begin{bmatrix} 1 & 2 & -1 & 3 \\ 2 & 4 & -4 & 7 \\ -1 & -2 & -1 & 1 \end{bmatrix}$

Here given,

$$A = \begin{bmatrix} 1 & 2 & -1 & 3 \\ 2 & 4 & -4 & 7 \\ -1 & -2 & -1 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 & 0 & 0 \\ 2 & 0 & -2 & 1 \\ -1 & 0 & -2 & 4 \end{bmatrix}$$

Apply,
$C_2' = C_2 - 2C_1$
$C_3' = C_3 + C_1$
$C_4' = C_4 - 3C_1$

$$= \begin{bmatrix} 1 & 0 & 0 & 0 \\ 2 & 0 & -2 & 1 \\ -3 & 0 & 0 & 3 \end{bmatrix}$$

Apply
$R_3' = R_3 - R_2$

$$= \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & -2 & 1 \\ 0 & 0 & 0 & 3 \end{bmatrix}$$

Apply $\boxed{R_2' = R_2 - 2R_1}$
$\cancel{C_3' = -\frac{C_3}{2}}$
$\cancel{R_3' = \frac{R_3}{3}}$
$\boxed{R_3' = R_3 + 3R_1}$

$$= \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Apply $C_3' = -\frac{C_3}{2}$
$R_3' = \frac{R_3}{3}$

$$= \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad [C_3' = C_3 - C_2]$$

$$= \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad [C_3 \rightleftarrows C_2]$$

$$= \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \quad [C_4 \rightleftarrows C_3]$$

$$= [I_3 \ \ 0]$$

Hence the rank of A is 3.

C.W

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Echelon form of a Matrix

Definition: If in a matrix,

(i) All the non-zero rows, if any, precede the zero rows.

(ii) The number of zero preceeding the first non-zero element in a row is less than the number of such zero in the succeeding row.

(iii) The first non-zero element in a row is unity,

then it is in the Echelon form.

Example

$$\begin{array}{l} \begin{bmatrix} 1 & 3 & 4 & 5 & 6 \\ 0 & 1 & 2 & 3 & 4 \\ 0 & 0 & 1 & 2 & 3 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \end{array} \begin{array}{l} \rightarrow \text{All full} \\ \rightarrow \text{1 এর পরে} \\ \rightarrow \text{1 থেকে শুরু} \\ \text{1st a zero} \end{array}$$

C.W

Example : 9(a) → page (62)

Find the ~~adjoint and~~ inverse of $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 2 \\ 3 & 3 & 4 \end{bmatrix}$

$$A = \begin{bmatrix} 1 & 2 & 1 \\ 3 & 2 & 3 \\ 1 & 1 & 2 \end{bmatrix}$$

$$\boxed{A^{-1} = \frac{Adj.\,of\,A}{|A|}}$$

Solution

Here given

$$A = \begin{bmatrix} 1 & 2 & 1 \\ 3 & 2 & 3 \\ 1 & 1 & 2 \end{bmatrix}$$

$$\therefore |A| = 1(4-3) - 2(6-3) + 1(3-2)$$

$$= 1 - 6 + 1$$

$$= -4 \neq 0$$

Now the cofactor of the matrix A are given by

$$C_{11} = \begin{vmatrix} 2 & 3 \\ 1 & 2 \end{vmatrix} \qquad C_{12} = -\begin{vmatrix} 3 & 3 \\ 1 & 2 \end{vmatrix} \qquad C_{13} = \begin{vmatrix} 3 & 2 \\ 1 & 1 \end{vmatrix}$$

$$= (4-3) \qquad = -(6-3) \qquad = (3-2)$$

$$= 1 \qquad = -3 \qquad = 1$$

$$C_{21} = -\begin{vmatrix} 2 & 1 \\ 1 & 2 \end{vmatrix} \qquad C_{22} = \begin{vmatrix} 1 & 1 \\ 1 & 2 \end{vmatrix} \qquad C_{23} = -\begin{vmatrix} 1 & 2 \\ 1 & 1 \end{vmatrix}$$

$$= -(4-1) \qquad = 2-1 \qquad = -(1-2)$$

$$= -3 \qquad = 1 \qquad = 1$$

$$C_{31} = \begin{vmatrix} 2 & 1 \\ 2 & 3 \end{vmatrix} \qquad C_{32} = -\begin{vmatrix} 1 & 1 \\ 3 & 3 \end{vmatrix} \qquad C_{33} = \begin{vmatrix} 1 & 2 \\ 3 & 2 \end{vmatrix}$$

$$= 6-2 \qquad = -(3-3) \qquad = (2-6)$$

$$= 4 \qquad = 0 \qquad = -4$$

So, the cofactor matrix of the matrix A is given by

$$A^c = \begin{bmatrix} 1 & -3 & 1 \\ -3 & 1 & 1 \\ 4 & 0 & -4 \end{bmatrix}$$

Hence the adjoint matrix of A is given by

$$\text{Adj of } A = (A^c)^t = \begin{bmatrix} 1 & -3 & 4 \\ -3 & 1 & 0 \\ 1 & 1 & -4 \end{bmatrix}$$

Therefore,

$$A^{-1} = \frac{\text{Adj. of } A}{|A|} = -\frac{1}{4}\begin{bmatrix} 1 & -3 & 4 \\ -3 & 1 & 0 \\ 1 & 1 & -4 \end{bmatrix}$$

$$= \begin{bmatrix} -1/4 & 3/4 & -1 \\ 3/4 & -1/4 & 0 \\ -1/4 & -1/4 & 1 \end{bmatrix} \text{ Ans.}$$

CamScanner

H.W

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Example - 6(a) Find adj of A and A^{-1}

when, $A = \begin{bmatrix} 1 & 3 & 3 \\ 1 & 4 & 3 \\ 1 & 3 & 4 \end{bmatrix}$

Solve

Here given,

$$A = \begin{bmatrix} 1 & 3 & 3 \\ 1 & 4 & 3 \\ 1 & 3 & 4 \end{bmatrix}$$

$$\therefore |A| = 1(16-9) - 3(4-3) + 3(3-4)$$

$$= 1 \neq 0$$

Now the cofactor of the matrix A are given by

$C_{11} = \begin{vmatrix} 4 & 3 \\ 3 & 4 \end{vmatrix}$, $C_{12} = -\begin{vmatrix} 1 & 3 \\ 1 & 4 \end{vmatrix}$, $C_{13} = \begin{vmatrix} 1 & 4 \\ 1 & 3 \end{vmatrix}$

$= 16-9$, $= -(4-3)$, $= 3-4$

$= 7$, $= -1$, $= -1$

$C_{21} = -\begin{vmatrix} 3 & 3 \\ 3 & 4 \end{vmatrix}$, $C_{22} = \begin{vmatrix} 1 & 3 \\ 1 & 4 \end{vmatrix}$, $C_{23} = -\begin{vmatrix} 1 & 3 \\ 1 & 3 \end{vmatrix}$

$= -3$, $= 4-3$, $= -(3-3)$

$= 1$, $= 0$

$C_{31} = \begin{vmatrix} 3 & 3 \\ 4 & 3 \end{vmatrix}$ $C_{32} = -\begin{vmatrix} 1 & 3 \\ 1 & 3 \end{vmatrix}$ $C_{33} = \begin{vmatrix} 1 & 3 \\ 1 & 4 \end{vmatrix}$

$= -3$ $= 0$ $= 1$

So the cofactor matrix of the matrix A is given by,

$$A^C = \begin{bmatrix} 7 & -1 & -1 \\ -3 & 1 & 0 \\ -3 & 0 & 1 \end{bmatrix}$$

Hence the adjoint matrix of A is given by

$$\text{Adj of } A = (A^C)^t = \begin{bmatrix} 7 & -3 & -3 \\ -1 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix}$$

Therefore, $A^{-1} = \dfrac{\text{Adj. of } A}{|A|}$

$$= \frac{1}{1} \times \begin{bmatrix} 7 & -3 & -3 \\ -1 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 7 & -3 & -3 \\ -1 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix}$$ Ans

Page (59)

Example - 7

If $A = \begin{bmatrix} 3 & -1 & 1 \\ -15 & 6 & -5 \\ 5 & -2 & 2 \end{bmatrix}$ find adj.A and A^{-1}

Solve

Here given, $A = \begin{bmatrix} 3 & -1 & 1 \\ -15 & 6 & -5 \\ 5 & -2 & 2 \end{bmatrix}$

$\therefore |A| = 3(12 - 10) + 1(-30 + 25) + 1(30 - 30)$

$= 1 \neq 0$

~~Also for the~~ Now the cofactor of the matrix A are given by,

$C_{11} = \begin{vmatrix} 6 & -5 \\ -2 & 2 \end{vmatrix}$; $C_{12} = -\begin{vmatrix} -15 & -5 \\ 5 & 2 \end{vmatrix}$; $C_{13} = \begin{vmatrix} -15 & 6 \\ 5 & -2 \end{vmatrix}$

$= 12 - 10$; $= -(-30 + 25)$; $= 30 - 30$

$= 2$; $= 5$; $= 0$

$C_{21} = -\begin{vmatrix} -1 & 1 \\ -2 & 2 \end{vmatrix}$; $C_{22} = \begin{vmatrix} 3 & 1 \\ 5 & 2 \end{vmatrix}$; $C_{23} = -\begin{vmatrix} 3 & -1 \\ 5 & -2 \end{vmatrix}$

$= -2+2$ $\qquad = 6-5$ $\qquad = -(-6+5)$

$= 0$ $\qquad = 1$ $\qquad = 1$

$C_{31} = \begin{vmatrix} -1 & 1 \\ 6 & -5 \end{vmatrix}$; $C_{32} = -\begin{vmatrix} 3 & 1 \\ -15 & -5 \end{vmatrix}$; $C_{33} = \begin{vmatrix} 3 & -1 \\ -15 & 6 \end{vmatrix}$

$= 3$

$= 0$

$= -1$

So the cofactor matrix of the matrix A is given by,

$$A^c = \begin{bmatrix} 2 & 5 & 0 \\ 0 & 1 & 1 \\ -1 & 0 & 3 \end{bmatrix}$$

Hence the adjoint matrix of A is given by

$$\text{Adj. of } A = (A^c)^t = \begin{bmatrix} 2 & 0 & -1 \\ 5 & 1 & 0 \\ 0 & 1 & 3 \end{bmatrix}$$

P-68 (16)
P-70 (18)
P-86 (3(b))

$\therefore$ Therefore, $A^{-1} = \dfrac{\text{Adj. of } A}{|A|} = \begin{bmatrix} 2 & 0 & -1 \\ 5 & 1 & 0 \\ 0 & 1 & 3 \end{bmatrix}$

Ans —

** Solution of linear Equation

◻ Solve the linear equations using matrix method, $x+y+z=6$, $x-y+z=2$, $2x+y-z=1$

Solve,

Given the linear equation,

$$x+y+z = 6$$
$$x-y+z = 2$$
$$2x+y-z = 1$$, body

Let, $A = \begin{vmatrix} 1 & 1 & 1 \\ 1 & -1 & 1 \\ 2 & 1 & -1 \end{vmatrix}$, $\cancel{X} = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$ → variable and

$$K = \begin{bmatrix} 6 \\ 2 \\ 1 \end{bmatrix} \longrightarrow \text{constant.}$$

Such that,

$$A\cancel{X} = K$$
$$\Rightarrow A^{-1}Ax = A^{-1}K$$
$$\Rightarrow \cancel{A} \cancel{X} = A^{-1}K \quad \text{—— (1)}$$

$AA^{-1} = A^{-1}A$
$= I$

Now, $|A| = \begin{vmatrix} 1 & 1 & 1 \\ 1 & -1 & 1 \\ 2 & 1 & -1 \end{vmatrix}$

$= 1(1-1) - 1(-1-2) + 1(1+2)$

$= 3+3 = 6$

Again the cofactor of the matrix of A is given by,

$C_{11} = \begin{vmatrix} -1 & 1 \\ 1 & -1 \end{vmatrix}$

$= 1-1 = 0$

$C_{12} = -\begin{vmatrix} 1 & 1 \\ 2 & -1 \end{vmatrix}$

$= -(-1-2) = 3$

$C_{13} = \begin{vmatrix} 1 & -1 \\ 2 & 1 \end{vmatrix}$

$= 1+2 = 3$

$C_{21} = -\begin{vmatrix} 1 & 1 \\ 1 & -1 \end{vmatrix}$

$= -(-1-1)$

$= 2$

$C_{22} = \begin{vmatrix} 1 & 1 \\ 2 & -1 \end{vmatrix}$

$= -1-2$

$= -3$

$C_{23} = -\begin{vmatrix} 1 & 1 \\ 2 & 1 \end{vmatrix}$

$= -(1-2)$

$= 1$

$C_{31} = \begin{vmatrix} 1 & 1 \\ -1 & 1 \end{vmatrix}$

$= 1+1$

$= 2$

$C_{32} = -\begin{vmatrix} 1 & 1 \\ 1 & 1 \end{vmatrix}$

$= -(1-1)$

$= 0$

$C_{33} = \begin{vmatrix} 1 & 1 \\ 1 & -1 \end{vmatrix}$

$= (-1-1)$

$= -2$

So, the cofactor matrix of A is given by,

$$A^c = \begin{bmatrix} 0 & 3 & 3 \\ 2 & -3 & 1 \\ 2 & 0 & -2 \end{bmatrix}$$

$\therefore$ Adj. of A is given by,

$$\text{Adj. of } A = (A^c)^t = \begin{bmatrix} 0 & 2 & 2 \\ 3 & -3 & 0 \\ 3 & 1 & -2 \end{bmatrix}$$

Hence,

$$A^{-1} = \frac{\text{Adj. of } A}{|A|}$$

$$= \frac{1}{6} \times \begin{bmatrix} 0 & 2 & 2 \\ 3 & -3 & 0 \\ 3 & 1 & -2 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & \frac{1}{3} & \frac{1}{3} \\ \frac{1}{2} & -\frac{1}{2} & 0 \\ \frac{1}{2} & \frac{1}{6} & -\frac{1}{3} \end{bmatrix}$$

Now from (1) we have,

$$M = A^{-1} K$$

$$= \begin{bmatrix} 0 & \frac{1}{3} & \frac{1}{3} \\ \frac{1}{2} & -\frac{1}{2} & 0 \\ \frac{1}{2} & \frac{1}{6} & -\frac{1}{3} \end{bmatrix} \times \begin{bmatrix} 6 \\ 2 \\ 1 \end{bmatrix}$$

$$X = \begin{bmatrix} 0 + \frac{1}{3}\times 2 + \frac{1}{3}\times 1 \\ \frac{1}{2}\times 6 - \frac{1}{2}\times 2 + 0 \\ \frac{1}{2}\times 6 + \frac{1}{6}\times 2 - \frac{1}{3} \end{bmatrix}$$

$$= \begin{bmatrix} 0 + \frac{2}{3} + \frac{1}{3} \\ 3 - 1 + 0 \\ 3 + \frac{1}{3} - \frac{1}{3} \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$$

$\therefore x = 1,\ y = 2,\ z = 3$ Ans —

$\boxed{\text{H.W}}$

<u>Page → 108</u>

<u>Example – 3(b)</u> Solve by matrix method,

$x + 2y + 3z = 14,\quad 3x + y + 2z = 11,$

$2x + 3y + z = 11$

Solution:

Here given equation are,

$$x + 2y + 3z = 14$$
$$3x + y + 2z = 11$$
$$2x + 3y + z = 11$$

Let, $A = \begin{bmatrix} 1 & 2 & 3 \\ 3 & 1 & 2 \\ 2 & 3 & 1 \end{bmatrix}$, $X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$, $K = \begin{bmatrix} 14 \\ 11 \\ 11 \end{bmatrix}$

Such that,

$$AX = K$$
$$\Rightarrow A^{-1}AX = A^{-1}K$$
$$\Rightarrow X = A^{-1}K \quad \text{—— (1)}$$

Now, $|A| = \begin{vmatrix} 1 & 2 & 3 \\ 3 & 1 & 2 \\ 2 & 3 & 1 \end{vmatrix}$

$$= 1(1-6) - 2(3-4) + 3(9-2)$$
$$= 18$$

Again the cofactor of the matrix of A is given by,

$C_{11} = \begin{vmatrix} 1 & 2 \\ 3 & 1 \end{vmatrix}$; $C_{12} = -\begin{vmatrix} 3 & 2 \\ 2 & 1 \end{vmatrix}$; $C_{13} = \begin{vmatrix} 3 & 1 \\ 2 & 3 \end{vmatrix}$

$= 1-6 = -5$; $= -(3-4) = 1$; $= 9-2 = 7$

$C_{21} = -\begin{vmatrix} 2 & 3 \\ 3 & 1 \end{vmatrix}$; $C_{22} = \begin{vmatrix} 1 & 3 \\ 2 & 1 \end{vmatrix}$; $C_{23} = -\begin{vmatrix} 1 & 2 \\ 2 & 3 \end{vmatrix}$

$= -(2-9) = 7$; $= 1-6 = -5$; $= -(3-4) = 1$

$C_{31} = \begin{vmatrix} 2 & 3 \\ 1 & 2 \end{vmatrix}$; $C_{32} = -\begin{vmatrix} 1 & 3 \\ 3 & 2 \end{vmatrix}$; $C_{33} = \begin{vmatrix} 1 & 2 \\ 3 & 1 \end{vmatrix}$

$= 4-3 = 1$; $= -(2-9) = 7$; $= 1-6 = -5$

So, the cofactor matrix of A is given by,

$$A^c = \begin{bmatrix} -5 & 1 & 7 \\ 7 & -5 & 1 \\ 1 & 7 & -5 \end{bmatrix}$$

$\therefore$ Adjoint of A is given by,

$$\text{Adj. of } A = (A^c)^T = \begin{bmatrix} -5 & 7 & 1 \\ 1 & -5 & 7 \\ 7 & 1 & -5 \end{bmatrix}$$

Hence, $A^{-1} = \dfrac{\text{Adj. of } A}{|A|}$

$$= \frac{1}{18} \times \begin{bmatrix} -5 & 7 & 1 \\ 1 & -5 & 7 \\ 7 & 1 & -5 \end{bmatrix}$$

$$= \begin{bmatrix} -\frac{5}{18} & \frac{7}{18} & \frac{1}{18} \\ \frac{1}{18} & -\frac{5}{18} & \frac{7}{18} \\ \frac{7}{18} & \frac{1}{18} & -\frac{5}{18} \end{bmatrix}$$

Now from (1) we have,

$$x = A^{-1} K$$

$$= \begin{bmatrix} -\frac{5}{18} & \frac{7}{18} & \frac{1}{18} \\ \frac{1}{18} & -\frac{5}{18} & \frac{7}{18} \\ \frac{7}{18} & \frac{1}{18} & -\frac{5}{18} \end{bmatrix} \times \begin{bmatrix} 14 \\ 11 \\ 11 \end{bmatrix}$$

$$= \begin{bmatrix} -\frac{5}{18} \times 14 + \frac{7}{18} \times 11 + \frac{1}{18} \times 11 \\ \frac{1}{18} \times 14 - \frac{5}{18} \times 11 + \frac{7}{18} \times 11 \\ \frac{7}{18} \times 14 + \frac{1}{18} \times 11 - \frac{5}{18} \times 11 \end{bmatrix}$$

$$X = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$$

$\therefore x = 1,\ y = 2,\ z = 3$. Ans—

Example - 4 (a) page (109)

Using matrix method, solve the following equations,

$2x - y + 3z = 9$, $x + y + z = 6$ and $x - y + z = 2$

Here given,

$$2x - y + 3z = 9$$
$$x + y + z = 6$$
$$x - y + z = 2$$

Let,

$$A = \begin{bmatrix} 2 & -1 & 3 \\ 1 & 1 & 1 \\ 1 & -1 & 1 \end{bmatrix},\quad X = \begin{bmatrix} x \\ y \\ z \end{bmatrix},\quad K = \begin{bmatrix} 9 \\ 6 \\ 2 \end{bmatrix}.$$

Such that, $AX = K$

$\Rightarrow A^{-1}AX = A^{-1}K \Rightarrow X = A^{-1}K$ —— (1)

Now, $|A| = \begin{vmatrix} 2 & -1 & 3 \\ 1 & 1 & 1 \\ 1 & -1 & 1 \end{vmatrix}$

$= 2(1+1) + 1(1-1) + 3\times(-1-1)$

$= 4 - 6 = -2$

Again the cofactor of the matrix of A is given by,

$C_{11} = \begin{vmatrix} 1 & 1 \\ -1 & 1 \end{vmatrix}$; $C_{12} = -\begin{vmatrix} 1 & 1 \\ 1 & 1 \end{vmatrix}$; $C_{13} = \begin{vmatrix} 1 & 1 \\ 1 & -1 \end{vmatrix}$

$= 1+1$ $\quad = -(1-1)$ $\quad = (-1-1) = -2$

$= 2$ $\quad = 0$

$C_{21} = -\begin{vmatrix} -1 & 3 \\ -1 & 1 \end{vmatrix}$; $C_{22} = \begin{vmatrix} 2 & 3 \\ 1 & 1 \end{vmatrix}$; $C_{23} = -\begin{vmatrix} 2 & -1 \\ 1 & -1 \end{vmatrix}$

$= -(-1+3)$ $\quad = 2-3$ $\quad = -(-2+1)$

$= -2$ $\quad = -1$ $\quad = 1$

$C_{31} = \begin{vmatrix} -1 & 3 \\ 1 & 1 \end{vmatrix}$; $C_{32} = -\begin{vmatrix} 2 & 3 \\ 1 & 1 \end{vmatrix}$; $C_{33} = \begin{vmatrix} 2 & -1 \\ 1 & 1 \end{vmatrix}$

$= -1-3$ $\quad = -(2-3)$ $\quad = 2+1$

$= -4$ $\quad = 1$ $\quad = 3$

So the cofactor matrix of A is given by,

$$A^C = \begin{bmatrix} 2 & 0 & -2 \\ -2 & -1 & 1 \\ -4 & 1 & 3 \end{bmatrix}$$

$\therefore$ Adjoint of A is given by,

$$\text{Adj of } A = (A^C)^T = \begin{bmatrix} 2 & -2 & -4 \\ 0 & -1 & 1 \\ -2 & 1 & 3 \end{bmatrix}$$

Hence, $A^{-1} = \dfrac{\text{Adj. of } A}{|A|} = \dfrac{1}{-2}\begin{bmatrix} 2 & -2 & -4 \\ 0 & -1 & 1 \\ -2 & 1 & 3 \end{bmatrix}$

$$= \begin{bmatrix} -1 & 1 & 2 \\ 0 & \frac{1}{2} & -\frac{1}{2} \\ 1 & -\frac{1}{2} & -\frac{3}{2} \end{bmatrix}$$

Now from (1) we have.

$$X = A^{-1}K$$

$$= \begin{bmatrix} -1 & 1 & 2 \\ 0 & \frac{1}{2} & -\frac{1}{2} \\ 1 & -\frac{1}{2} & -\frac{3}{2} \end{bmatrix} \times \begin{bmatrix} 9 \\ 6 \\ 2 \end{bmatrix}$$

$$= \begin{bmatrix} -9+6+4 \\ 0+3-1 \\ 9-3-3 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$$

$\therefore x = 1, y = 2, z = 3$

Ans -

Example – 4 (b)   solve the following equations by matrix method

Here given,

$$x + 2y + z = 2$$
$$3x + 5y + 5z = 4$$
$$2x + 4y + 3z = 3$$

Ans –
$x = 3,\ y = 0,\ z =$

Let, $A = \begin{bmatrix} 1 & 2 & 1 \\ 3 & 5 & 5 \\ 2 & 4 & 3 \end{bmatrix}$, $X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$, $K = \begin{bmatrix} 2 \\ 4 \\ 3 \end{bmatrix}$

Such that,

$$AX = K$$
$$A^{-1}AX = A^{-1}K$$
$$X = A^{-1}K \longrightarrow (1)$$

Characteristic Equation of a matrix

(i) Characteristic Matrix of A :

The matrix $A - \lambda I$ is called the characteristic matrix of A, where I is the identity matrix

(ii) Characteristic polynomial of A :

The determinant $|A - \lambda I|$ is called the characteristic ~~equation~~ polynomial of A ~~and~~ its roots ~~are~~ called the characteristic

(iii) Characteristic equation of A : The equation $|A - \lambda I| = 0$ is known as the characteristic equation of A and its roots are called the characteristic roots or latent roots or characteristic or proper values of A.

(iv) Spectrum of A : The set of all eigen values of the matrix A is called the spectrum of A. $\boxed{A : \lambda = \{1, 3\}}$

(v) Eigen values problems : The problem of finding the eigen values of a matrix is known as the eigen value problem. GA

① $A = \begin{bmatrix} 1-\lambda & 2 & 3 \\ 4 & 5-\lambda & 6 \\ 7 & 8 & 9-\lambda \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

$= \lambda^3 - 3\lambda^2 + 4\lambda + 5$

State Cayley Hamilton's (C-H) theorem and verify the theorem for the matrix.

$A = \begin{bmatrix} 2 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix}$ and hence find A^{-1}

Statement : Every square matrix satisfies its characteristic equation. OR,

If $|A - \lambda I| = (-1)^n [\lambda^n + a_1\lambda^{n-1} + a_2\lambda^{n-2} + \cdots + a_n]$ be the characteristic polynomial of $n \times n$ matrix $A = [a_{ij}]$, then the matrix equation

$X_n + a_1 X^{n-1} + \cdots + a_n I = 0$ is satisfied by $X = A$

$\boxed{A^n + a_1 A^{n-1} + \cdots + a_n I = 0}$

Given that,

$$A = \begin{bmatrix} 2 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix}$$

Now,

$$A - \lambda I = \begin{bmatrix} 2 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 2 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix} - \begin{bmatrix} \lambda & 0 & 0 \\ 0 & \lambda & 0 \\ 0 & 0 & \lambda \end{bmatrix}$$

$$= \begin{bmatrix} 2-\lambda & -1 & -1 \\ -1 & 2-\lambda & -1 \\ 1 & -1 & 2-\lambda \end{bmatrix}$$

$\therefore$ The characteristic equation of A,

$$|A - \lambda I| = 0$$

$$\Rightarrow \begin{vmatrix} 2-\lambda & -1 & 1 \\ -1 & 2-\lambda & -1 \\ 1 & -1 & 2-\lambda \end{vmatrix} = 0$$

$$\Rightarrow 2-\lambda \{ (2-\lambda) + 1 \} + 1 (-2 + \lambda + 1) + 1 (1 - 2 + \lambda)$$

$$\boxed{(a-b)^3 = a^3 - 3a^2b + 3ab^2 - b^3}$$

$$= (2-\lambda)^3 + 2 - \lambda + 2 + \lambda + 1 + 1 - 2 + \lambda$$

$$= 8 - 3 \times 4 \cdot \lambda + 3 \cdot 2 \cdot \lambda^2 - \lambda^3 + 2 - \lambda + 2 + \lambda + 1 + 1 - 2 + \lambda$$

$\Rightarrow$ ~~$12\lambda + 6\lambda^2 + \lambda^3 + \lambda + 12 = 0$~~

$\therefore$ The characteristic equation of A,

$$|A - \lambda I| = 0$$

$$\Rightarrow \begin{vmatrix} 2-\lambda & -1 & 1 \\ -1 & 2-\lambda & -1 \\ 1 & -1 & 2-\lambda \end{vmatrix} = 0$$

$$\Rightarrow 2-\lambda \{(2-\lambda)\times(2-\lambda) - 1\} + 1(-2+\lambda+1) + 1(1-2+\lambda) = 0$$

$$\Rightarrow 2-\lambda\{(4-2\lambda-2\lambda+\lambda^2)-1\} + (-2+\lambda+1 \; 1-2+\lambda) = 0$$

$$\Rightarrow 8 - 4\lambda - 4\lambda + 2\lambda^2 - 4\lambda + 2\lambda^2 + 2\lambda^2 - \lambda^3 - 2 + \lambda - 2 + 2\lambda = 0$$

$$\Rightarrow -\lambda^3 + 6\lambda^2 - 9\lambda + 4 = 0$$

$$\Rightarrow \lambda^3 - 6\lambda^2 + 9\lambda - 4 = 0$$

Now by the caley Hamilton's theorem, we have,

$$A^3 - 6A^2 + 9A - 4 = 0 \quad \text{--- (1)}$$

Now, $A^2 = A \times A$

$$= \begin{bmatrix} 2 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix} \times \begin{bmatrix} 2 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix}$$

$$= \cancel{4-2+2 \quad 1-2+1}$$

$$= \begin{bmatrix} 4+1+1 & -2-2-1 & 2+1+2 \\ -2-2-1 & 1+4+1 & -1-2-2 \\ 2+1+2 & -1-2-2 & 1+1+4 \end{bmatrix}$$

$$= \begin{bmatrix} 6 & -5 & 5 \\ -5 & 6 & -5 \\ 5 & -5 & 6 \end{bmatrix}$$

also, $A^3 = A^2 \times A$

$$= \begin{bmatrix} 6 & -5 & 5 \\ -5 & 6 & -5 \\ 5 & -5 & 6 \end{bmatrix} \times \begin{bmatrix} 2 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix}$$

$$= \begin{bmatrix} 12+5+5 & -6-10-5 & 6+5+10 \\ -10-6-5 & 5+12+5 & -5-6-10 \\ 10+5+6 & -5-10-6 & 5+5+12 \end{bmatrix}$$

$$= \begin{bmatrix} 22 & -21 & 21 \\ -21 & 22 & -21 \\ 21 & -21 & 22 \end{bmatrix}$$

Now from (1) we have,

$$L.S = A^3 - 6A^2 + 9A - 4$$

$$= \begin{bmatrix} 22 & -21 & 21 \\ -21 & 22 & -21 \\ 21 & -21 & 22 \end{bmatrix} - 6\begin{bmatrix} 6 & -5 & 5 \\ -5 & 6 & -5 \\ 5 & -5 & 6 \end{bmatrix} + 9\begin{bmatrix} 2 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix} - 4\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 22 & -21 & 21 \\ -21 & 22 & -21 \\ 21 & -21 & 22 \end{bmatrix} - \begin{bmatrix} 36 & -30 & 30 \\ -30 & 36 & -30 \\ 30 & -30 & 36 \end{bmatrix} + \begin{bmatrix} 18 & -9 & 9 \\ -9 & 18 & -9 \\ 9 & -9 & 18 \end{bmatrix} - \begin{bmatrix} 4 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 4 \end{bmatrix}$$

$$= \begin{bmatrix} 40 & -30 & 30 \\ -30 & 40 & -30 \\ 30 & -30 & 40 \end{bmatrix} - \begin{bmatrix} 40 & -30 & 30 \\ -30 & 40 & -30 \\ 30 & -30 & 40 \end{bmatrix}$$

$= 0$ Hence Clay Hamilton's theorem verified.

$= R.S$

=

$A^3-6A^2+9A-4=0$ $\cancel{AO}$ → [illegible],

A^{-2} [illegible]

3rd part :

We have,

$$A^3-6A^2-9A-4=0$$

Now multiplying by A^{-1} we get,

$$A^3 \cdot A^{-1} - 6A^2 \cdot A^{-1} - 9A \cdot A^{-1} - 4 \times A^{-1} = 0$$

$$\Rightarrow A^2 - 6A + 9 - 4A^{-1} = 0$$

$$\Rightarrow 4A^{-1} = A^2 - 6A + 9$$

$$= \begin{bmatrix} 6 & -5 & 5 \\ -5 & 6 & -5 \\ 5 & -5 & 6 \end{bmatrix} - 6\begin{bmatrix} 2 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix} + 9\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 6 & -5 & 5 \\ \cancel{-5} & 6 & -5 \\ 5 & -5 & 6 \end{bmatrix} - \begin{bmatrix} 12 & -6 & 6 \\ -6 & 12 & -6 \\ 6 & -6 & 12 \end{bmatrix} + \begin{bmatrix} 9 & 0 & 0 \\ 0 & 9 & 0 \\ 0 & 0 & 9 \end{bmatrix}$$

$$= \begin{bmatrix} 15 & -5 & 5 \\ -5 & 15 & -5 \\ 5 & -5 & 15 \end{bmatrix} - \begin{bmatrix} 12 & -6 & 6 \\ -6 & 12 & -6 \\ 6 & -6 & 12 \end{bmatrix}$$

$$\Rightarrow 4A^{-1} = \begin{bmatrix} 3 & 1 & -1 \\ 1 & 3 & 1 \\ -1 & & 3 \end{bmatrix}$$

$$\Rightarrow A^{-1} = \frac{1}{4}\begin{bmatrix} 3 & 1 & -1 \\ 1 & 3 & 1 \\ -1 & 1 & 3 \end{bmatrix}$$ Ans —

Hence, $$A^{-1} = \begin{bmatrix} 3/4 & 1/4 & -1/4 \\ 1/4 & 3/4 & 1/4 \\ -1/4 & 1/4 & 3/4 \end{bmatrix}$$ Ans —

H.W

P(183) **Example — 10** Verify cayley Hamilton's theorem for the matrix $A = \begin{bmatrix} 1 & 0 & 2 \\ 0 & 2 & 1 \\ 2 & 0 & 3 \end{bmatrix}$. Hen

of otherwise compute A^{-1}

Given that,

$$A = \begin{bmatrix} 1 & 0 & 2 \\ 0 & 2 & 1 \\ 2 & 0 & 3 \end{bmatrix}.$$

Now, $A - \lambda I = \begin{bmatrix} 1 & 0 & 2 \\ 0 & 2 & 1 \\ 2 & 0 & 3 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

$= \begin{bmatrix} 1 & 0 & 2 \\ 0 & 2 & 1 \\ 2 & 0 & 3 \end{bmatrix} - \begin{bmatrix} \lambda & 0 & 0 \\ 0 & \lambda & 0 \\ 0 & 0 & \lambda \end{bmatrix}$

$= \begin{bmatrix} 1-\lambda & 0 & 2 \\ 0 & 2-\lambda & 1 \\ 2 & 0 & 3-\lambda \end{bmatrix}$

$\therefore$ The characteristic equation of A,

$|A - \lambda I| = 0$

$\Rightarrow \begin{vmatrix} 1-\lambda & 0 & 2 \\ 0 & 2-\lambda & 1 \\ 2 & 0 & 3-\lambda \end{vmatrix} = 0$

$\Rightarrow 1-\lambda\{(2-\lambda)\times(3-\lambda) - \} - 0 + 2(0 - 4 + 2\lambda) = 0$

$\Rightarrow 1-\lambda(6 - 2\lambda - 3\lambda + \lambda^2) + 0 - 8 + 4\lambda = 0$

$\Rightarrow 1-\lambda(6 - 5\lambda + \lambda^2) - 6 + 4\lambda = 0$

$\Rightarrow 6 - 5\lambda + \lambda^2 - 6\lambda + 5\lambda^2 - \lambda^3 - 8 + 4\lambda = 0$

$\Rightarrow -\lambda^3 + 6\lambda^2 - 7\lambda - 2 = 0$

$\Rightarrow \lambda^3 - 6\lambda^2 + 7\lambda + 2 = 0$

Now by the caley Hamilton's theorem we have,

$$A^3 - 6A^2 + 7A + 2 = 0 \text{ —— } (1)$$

Now, $A^2 = A \times A$

$$= \begin{bmatrix} 1 & 0 & 2 \\ 0 & 2 & 1 \\ 2 & 0 & 3 \end{bmatrix} \times \begin{bmatrix} 1 & 0 & 2 \\ 0 & 2 & 1 \\ 2 & 0 & 3 \end{bmatrix}$$

$$= \begin{bmatrix} 1+0+4 & 0+0+0 & 2+0+6 \\ 0+0+2 & 0+4+0 & 0+2+3 \\ 2+0+6 & 0+0+0 & 4+0+9 \end{bmatrix}$$

$$= \begin{bmatrix} 5 & 0 & 8 \\ 2 & 4 & 5 \\ 8 & 0 & 13 \end{bmatrix}$$

$A^3 = A^2 \times A$

$$= \begin{bmatrix} 5 & 0 & 8 \\ 2 & 4 & 5 \\ 8 & 0 & 13 \end{bmatrix} \times \begin{bmatrix} 1 & 0 & 2 \\ 0 & 2 & 1 \\ 2 & 0 & 3 \end{bmatrix}$$

$$= \begin{bmatrix} 21 & 0 & 34 \\ 12 & 8 & 23 \\ 34 & 0 & 55 \end{bmatrix}$$

also, ~~A^3~~ Now from (i) we have,

$$L.S = A^3 - 6A^2 + 7A + 2$$

$$\to \begin{bmatrix} 21 & 0 & 34 \\ 12 & 8 & 23 \\ 34 & 0 & 55 \end{bmatrix} - 6\begin{bmatrix} 5 & 0 & 8 \\ 2 & 4 & 5 \\ 8 & 0 & 13 \end{bmatrix} + 7\begin{bmatrix} 1 & 0 & 2 \\ 0 & 2 & 1 \\ 2 & 0 & 3 \end{bmatrix} + 2\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 21 & 0 & 34 \\ 12 & 8 & 23 \\ 34 & 0 & 55 \end{bmatrix} + \begin{bmatrix} -30 & 0 & -48 \\ -12 & 24 & -30 \\ -48 & 0 & -78 \end{bmatrix} + \begin{bmatrix} 7 & 0 & 14 \\ 0 & 14 & 7 \\ 14 & 0 & 21 \end{bmatrix} + \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$= O$, ~~where O is~~ Hence caly Hamilton's theorom verified

$= R.S$

we have,

$$A^3 - 6A^2 + 7A + 2 = 0$$

Now multiplying by A^{-1} we get

$$A^3 . A^{-1} - 6A^2 . A^{-1} + 7A . A^{-1} + 2A^{-1} = 0$$

$$\Rightarrow A^2 - 6A + 7 + 2A^{-1} = 0$$

$$\Rightarrow 2A^{-1} = -A^2 + 6A - 7$$

$$= -\begin{bmatrix} 5 & 0 & 8 \\ 2 & 4 & 5 \\ 8 & 0 & 13 \end{bmatrix} + 6\begin{bmatrix} 1 & 0 & 2 \\ 0 & 2 & 1 \\ 2 & 0 & 3 \end{bmatrix} - 7\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} -5 & 0 & -8 \\ -2 & -4 & -5 \\ -8 & 0 & -13 \end{bmatrix} + \begin{bmatrix} 6 & 0 & 12 \\ 0 & 12 & 6 \\ 12 & 0 & 18 \end{bmatrix} + \begin{bmatrix} -7 & 0 & 0 \\ 0 & -7 & 0 \\ 0 & 0 & -7 \end{bmatrix}$$

$$2A^{-1} = \begin{bmatrix} -6 & 0 & 4 \\ -2 & 1 & 1 \\ 4 & 0 & -2 \end{bmatrix}$$

$$\Rightarrow A^{-1} = \frac{1}{2}\begin{bmatrix} -6 & 0 & 4 \\ -2 & 1 & 1 \\ 4 & 0 & -2 \end{bmatrix}$$ Ans-

Hence,

$$A^{-1} = \begin{bmatrix} -3 & 0 & 2 \\ -1 & \frac{1}{2} & 1/2 \\ 2 & 0 & -1 \end{bmatrix} \text{ Ans}$$

Q.

Example – 15 (a) (page 189)

using cayley Hamilton theorem calculate $2A^5 + 3A^4 + A^2 - 11I$, where, $A = \begin{bmatrix} 3 & 1 \\ -1 & 2 \end{bmatrix}$

Solution

Given that, $A = \begin{bmatrix} 3 & 1 \\ -1 & 2 \end{bmatrix}$

Now, $A - \lambda I = \begin{bmatrix} 3 & 1 \\ -1 & 2 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

$= \begin{bmatrix} 3 & 1 \\ -1 & 2 \end{bmatrix} - \begin{bmatrix} \lambda & 0 & 0 \\ 0 & \lambda & 0 \\ 0 & 0 & \lambda \end{bmatrix}$

$= \begin{bmatrix} 3-\lambda & 1 \\ -1 & 2-\lambda \end{bmatrix}$

$\therefore$ The characteristic equation of A,

$$|A-\lambda I| = 0$$

$$\Rightarrow \begin{vmatrix} 3-\lambda & 1 \\ -1 & 2-\lambda \end{vmatrix} = 0$$

$$\Rightarrow (3-\lambda)\times(2-\lambda)+1 = 0$$

$$\Rightarrow 6-3\lambda-2\lambda+\lambda^2+1 = 0$$

$$\Rightarrow \lambda^2-5\lambda+7 = 0$$

Now by the Caley Hamilton's theorem, we have

$$A^2-5A+7=0 \quad (1)$$

Now, $A^2 = \begin{bmatrix} 3 & 1 \\ -1 & 2 \end{bmatrix} \times \begin{bmatrix} 3 & 1 \\ -1 & 2 \end{bmatrix}$

$$= \begin{bmatrix} 8 & 5 \\ -5 & 3 \end{bmatrix}$$

Now from (1) we have

$$L.S = A^2-5A+7$$

$$= \begin{bmatrix} 8 & 5 \\ -5 & 3 \end{bmatrix} - 5\begin{bmatrix} 3 & \\ -1 & \end{bmatrix}$$

$$+7\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 8 & 5 \\ -5 & 3 \end{bmatrix} + \begin{bmatrix} -15 & -5 \\ 5 & -10 \end{bmatrix} + \begin{bmatrix} 7 & 0 & 0 \\ 0 & 7 & 0 \\ 0 & 0 & 7 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

$= O$

$=$ R.S Hence caley Hamilton's theorom verified.

Now given

$$A^2 - 5A + 7I = 0 \quad \text{—— (1)}$$

$\therefore \; 2A^5 + 3A^4 + A^2 - 11I$

$= 2A^3(A^2 - 5A + 7I) + 13A^4 - 14A^3 - A^2 - 11I$

$= 2A^3(A^2 - 5A + 7I) + 51A^3$

$= 2A^3(O) + 13A^2(A^2 - 5A + 7I) - 92A^2 - 11I \rightarrow$ from (1)

$= 13A^2(0) + 51(A^2 - 5A + 7I) + 163A^2 - 355A - 11I$

$= 5A(0) + 163(A^2 - 5A + 7I) + 460A - 1152I$

$= 163(0) + 460A - 1152I$

$- 460A - 1152I$

$$= 460\begin{bmatrix} 3 & 1 \\ -1 & 2 \end{bmatrix} - \begin{bmatrix} 1152 & 0 \\ 0 & 1152 \end{bmatrix}$$

$$= \begin{bmatrix} 1380 & 460 \\ -460 & 920 \end{bmatrix} - \begin{bmatrix} 1152 & 0 \\ 0 & 1152 \end{bmatrix}$$

$$= \begin{bmatrix} 228 & 460 \\ -460 & -232 \end{bmatrix}$$ Ans

System of Linear Equation

Example - 4

A manufacturer produces three products: Toothpaste, soap and shampoo which sell in the market. Annual sale volumes are indicated as follows.

Market	Products		
	Toothpaste	Soap	Shampoo
New Market	8000	10000	15000
Shopping complex	10000	2000	20000

(i) If unit (mini packet) sale prices of Toothpaste, soap and shampoo are tk 2.225 tk 3.50 and tk 3.00 respectively. Find the total revenue in each market with the help of matrix

(ii) If the unit costs of the above three products are tk 1.60 tk, 1.90 and tk 2.40 respectively. Find gross profit with the help of matrices.

Solution.

The product matrix gives the total revenue in each market.

Let, Unit sale price of three products be

$$A = \begin{pmatrix} 2.25 & 3.50 & 3.00 \end{pmatrix}$$

Annual sale of three goods in two markets be, $B = \begin{pmatrix} 8000 & 10000 \\ 10000 & 2000 \\ 15000 & 20000 \end{pmatrix}$

Now,

$$A \times B = \begin{pmatrix} 2.25 & 3.50 & 3.00 \end{pmatrix} \times \begin{pmatrix} 8000 & 10000 \\ 10000 & 2000 \\ 15000 & 20000 \end{pmatrix}$$

$$= \begin{pmatrix} 2.25 \times 8000 + 3.50 \times 10000 + 3 \times 15000 & 2.25 \times 10000 + 3.50 \times 2000 + 3 \times 20000 \end{pmatrix}$$

$$= \begin{pmatrix} 98000 & 89500 \end{pmatrix}$$

Total revenue from the new market Tk. 98,000

Total " " the shopping complex Tk. 89,500

$\therefore$ Total revenue from two markets: TK. 1,87,500/-

(ii) Similarly the costs of products with manufactures sells in the market are,

$$A = \begin{pmatrix} 1.60 & 1.90 & 2.40 \end{pmatrix},$$

$$B = \begin{pmatrix} 8000 & 10000 \\ 10000 & 2000 \\ 15000 & 20000 \end{pmatrix}$$

$$\therefore A \times B = \begin{pmatrix} 1.60 & 1.90 & 2.40 \end{pmatrix} \times \begin{pmatrix} 8000 & 10000 \\ 10000 & 2000 \\ 15000 & 20000 \end{pmatrix}$$

$$= (1.60 \times 8000 + 1.90 \times 10000 + 2.40 \times 15000 \quad 1.60 \times 10000 + 1.90 \times 2000 + 2.40 \times 20000)$$

$$= \begin{pmatrix} 67800 & 67800 \end{pmatrix}$$

$\therefore$ Total cost of products for new market tk 67,800

$\therefore$ Total " " " " shopping complex TK 67800

$\therefore$ Total = TK. 1,35,600

So, revined gross profit =

(Total revenue − total cost)

$= (187500 - 135600)$

$=$ Tk. 51,900

Ans —

System of linear Equation

① **Homogenous system of linear equation :** The system is said to be homogeneous if the constants $b_1, b_2, b_3 \ldots b_m$ are zero $[b_1, b_2 = 0]$

② **Non-homogeneous system of linear equation :** The system is said to be non-homogenous if the constants $b_1, b_2, b_3 \ldots b_m$ are all 2 non-zero.

③ **Inconsistent system of linear equation :** If the given system has no rceal solution then it is called incosistent system of linear equation. (Solution হবে না)

④ **Consistent system of linear equation :** If the given system has a solution then the system ō said to be consistent system of linear equation.

(v) No solution system of linear equation

(vi) unique solution system of linear equation

(vii) More than one solution " " "

(viii) Zero solution " " " "

Formula : $L_i = -a_{i1}L_1 + a_{11}L_i$ where $i > 1$

Example - 16 : Solve the system

$2x + y - 2z = 10$ ---- L_1

$3x + 2y + 2z = 1$ ---- L_2

$5x + 4y + 3z = 4$ ---- L_3

Solution Given the system of linear equation

$2x + y - 2z = 10$ ---- L_1

$3x + 2y + 2z = 1$ ---- L_2

$5x + 4y + 3z = 4$ ---- L_3

We know that,

$$L_i = -a_{i1}L_1 + a_{11}L_i, \quad (i>1) \qquad (1)$$

Now putting $i=2$, in equation (1) we get,

$$L_2 = -a_{21}L_1 + a_{11}L_2$$

$$= -3(2x+y-2z-10) + 2(3x+2y+2z-1)$$

$$= -6x - 3y + 6z + 30 + 6x + 4y + 4z - 2$$

$$= y + 10z + 28$$

Again putting $i=3$, in equation (1) we get,

$$L_3 = -a_{31}L_1 + a_{11}L_3$$

$$= -5(2x+y-2z-10) + 2(5x+4y+3z-4)$$

$$= -\cancel{10x} - 5y + 10z + 50 + \cancel{10x} + 8y + 6z - 8$$

$$= 3y + 16z + 42$$

Thus we obtain the following new system,

$$2x + y - 2z = 10$$

$$y + 10z = -28 \text{ ——— } L_1$$

$$3y + 16z = -42 \text{ ——— } L_2$$

Now putting,

$i = 2$ in equation (1) we get,

$$L_2 = -a_{21}L_1 + a_{11}L_2$$

$$= -3(y + 10z + 28) + 1(3y + 16z + 42)$$

$$= -3y - 30z - 84 + 3y + 16z + 42$$

$$= -14z - 42$$

Thus we obtain the following system,

$$2x + y - 2z = 10$$

$$y + 10z = -28$$

$$-14z = 42$$

Now, $-14z = 42$

$\therefore z = -3$

Again, $y + 10z = -28$

$\Rightarrow y + 10(-3) = -28$

$\Rightarrow y = -28 + 30$

$\therefore y = 2$

Again,

$2x + y - 2z = 0$

$\Rightarrow 2x + 2 - 2(-3) = 10$

$\Rightarrow 2x + 2 + 6 = 10$

$\Rightarrow 2x = 2$

$\Rightarrow x = 1$

Ans : $(x, y, z) = (1, 2, 3)$

or, $x = 1$

$y = 2$

$z = 3$

Ex (Sheet)

17 $\rightarrow$ P(44)

18 $\rightarrow$ (45)

19 –

20

21

22

23

Example - 18

Solve the system,

$$2x + 3y - 2z = 5 \text{ ------ } L_1$$
$$x - 2y + 3z = 2 \text{ ------ } L_2$$
$$4x - y + 4z = 1 \text{ ------ } L_3$$

Solution

We know that,

$$L_i = -a_{i1}L_1 + a_{11}L_i, \quad (i > 1) \text{ --- } (1)$$

Now putting $i = 2$, in equation (1) we get,

$$L_2 = -a_{21}L_1 + a_{11}L_2$$
$$= -1(2x + 3y - 2z) + 2(x - 2y + 3z - 2)$$
$$= -\cancel{2x} - 3y + 2z + \cancel{2x} - 4y + 6z - 4$$
$$= -7y + 8z + 1$$

Again putting $i = 3$ in equation (1) we get,

$$L_3 = -a_{31}L_1 + a_{11}L_3$$
$$= -4(2x+3y-2z-5) + 2(4x-y+4z-1)$$
$$= -8x - 12y + 8z + 20 + 8x - 2y + 8z - 2$$
$$= -14y + 16z + 18$$

Thus we obtain the following new system,

$$2x + 3y - 2z = 5$$
$$-7y + 8z = -1 \text{ ——— } L_1$$
$$-14y + 16z = -18 \text{ ——— } L_2$$

Now putting, $i=2$ in eqation ① we get,

$$L_2 = -a_{21}L_1 + a_{11}L_2$$
$$= -(-14)(-7y+8z+1) + -7(-14y+16z+18)$$
$$= -98y + 112z + 14 + 98y - 112z - 126$$
$$= -112$$

Thus we obtain the following system

$$2x + 3y - 2z = 5$$
$$-7y + 8z = -1$$
$$0 = 112$$

It is not possible.

Hence, the system is inconsistent and has no solution

(15) Solve the system,

$$x + 2y - 3z = 6$$
$$2x - y + 4z = 2$$
$$4x + 3y - 2z = 14$$

Solve

We know that,

$$L_i = -a_{i1}L_1 + a_{11}L_i \quad (i > 1) \quad \text{—— (1)}$$

Now putting $i = 2$ in equation (1) we get,

$$L_2 = -a_{21}L_1 + a_{11}L_2$$
$$= -2(x + 2y - 3z - 6) + 1(2x - y + 4z - 2)$$
$$= -2x - 4y + 6z + 12 + 2x - y + 4z - 2$$
$$= -5y + 10z + 10$$

Again putting $i = 3$, in equation (1) we get,

$$L_3 = -a_{31}L_1 + a_{11}L_3$$
$$= -4(x + 2y - 3z - 6) + 1(4x + 3y - 2z - 14)$$
$$= -4x - 8y + 12z + 24 + 4x + 3y - 2z - 14$$
$$= -5y + 10z + 10$$

Thus we obtain the following system,

$$x + 2y - 3z = 6$$
$$-5y + 10z = -10 \text{ ——— } L_1$$
$$-5y + 10z = -10 \text{ ——— } L_2$$

Now putting $i = 2$ in equation (1) we get,

$$L_2 = -a_{21} L_1 + a_{11} L_2$$
$$= -(-5)(-5y + 10z + 10) + (-5)(-5y + 10z + 10)$$
$$= -25y + 50z + 50 + 25y - 50z - 5$$
$$= 0$$

Thus we obtain the following system,

$$x + 2y - 3z = 6$$
$$-5y + 10z = -10$$
$$0 = 0$$

Hence in echelon form, there are only two equations in three unknowns, then th system has a non-zero solution and in particular $(3-2) = 1$ free variable, which is z.

Let, $z = 1$

$\Rightarrow \cancel{x} - 5y + 10z = -10$

$\Rightarrow -5y + 10 = -10$

$\Rightarrow -5y = -20$

$\therefore y = 4$

Again,

$x + 2y - 3z = 6$

$\Rightarrow x + 2\times2 - 3\times1 = 6$

$\Rightarrow x + 8 - 3 = 6$

$\Rightarrow x + 5 = 6$

$\therefore x = 1$

$\therefore (x, y, z) = (1, 4, 1)$

Example : 20

Solve the system,

$$x + 2y - z = 0$$
$$2x + 5y + 2z = 0$$
$$x + 4y + 7z = 0$$
$$x + 3y + 3z = 0$$

solve

~~Here~~ giver

Here given,

$$x + 2y - z = 0 \quad \text{——} \quad L_1$$

$$2x + 5y + 2z = 0 \quad \text{——} \quad L_2$$

$$x + 4y + 7z = 0 \quad \text{——} \quad L_3$$

$$x + 3y + 3z = 0 \quad \text{——} \quad L_4$$

We know,

$$L_i = -a_{i1} L_1 + a_{11} L_i \,, \quad (i > 1) \quad \text{——} \ (1)$$

Now putting $i = 2$ in eq^n (1),

$$L_2 = -a_{21} L_1 + a_{11} L_2$$

$$= -2(x + 2y - z) + 1(2x + 5y + 2z)$$

$$= -2x - 2y + 2z + 2x + 5y + 2z$$

$$= y + 4z$$

Now putting $i = 3$ in eq^n (1),

$$L_3 = -a_{31} L_1 + a_{11} L_3$$

$$= -1(x + 2y - z) + 1(x + 4y + 7z)$$

$$= -x - 2y + z + x + 4y + 7z$$

$$= 2y + 8z$$

$i = 4$,

$$L_4 = -a_{41}L_1 + a_{11}L_4$$
$$= -1(x+2y-z) + 1(x+3y+3z)$$
$$= -x - 2y + z + x + 3y + 3z$$
$$= y + 4z$$

Then we obtain the system of equations,

$$x + 2y - z = 0$$
$$y + 4z = 0$$
$$2y + 8z = 0 \Rightarrow y + 4z = 0 \quad [\text{Dividing by } 2]$$
$$y + 4z = 0$$

Since the 2nd, 3rd & 4th equations are identical, we can disregard one of them.

$$x + 2y - z = 0$$
$$y + 4z = 0$$

In echelon form, there are only two equations in three unknowns, then the system has a non-zero solution and in particular $(3-2)=1$ free variable whi

is z.

Let $z=1$

$y+4z=0$

$\Rightarrow y+4\times 1=0$

$\therefore y=-4$

Again, $x+2\times(-4)-1=0$

$\Rightarrow x-9=0$

$\therefore x=9$

$\therefore (x,y,z)=(9,-4,1)$

Example: 21

Solve the system:

$$x + y - z = 0$$
$$2x + 4y - z = 0$$
$$3x + 2y + 2z = 0$$

Solve

Here given,

$$x + y - z = 0 \text{ —— } L_1$$
$$3x + 4y - z = 0 \text{ —— } L_2$$
$$3x + 2y + 2z = 0 \text{ —— } L_3$$

We know that,

$$L_i = -a_{i1}L_1 + a_{11}L_i, \ (i > 1) \text{ —— } (1)$$

Now putting $i = 2$ in equation (1) we get,

$$L_2 = -a_{21}L_1 + a_{11}L_2$$
$$= -2(x + y - z) + 1(2x + 4y - z)$$
$$= -2x - 2y + 2z + 2x + 4y - z$$
$$= 2y + z$$

$i = 3$,

$$L_3 = -a_{31}L_1 + a_{11}L_3$$
$$= -3(x + y - z) + 1(3x + 2y + 2z)$$
$$= -3x - 3y + 3z + 3x + 2y + 2z$$
$$= -y + 5z$$

Then we obtain the system of equations -

$$x + y - z = 0$$
$$2y + z = 0 \quad \text{———} \; L_1$$
$$-y + 5z = 0 \quad \text{———} \; L_2$$

Again putting $i = 2$ in eqn (1) we get,

$$L_2 = -a_{21}L_1 + a_{11}L_2$$
$$= 1(2y + z) + 2(-y + 5z)$$
$$= 2y + z - 2y + 10z$$
$$= 11z$$

Then we obtain the system of equations,

$$x + y - z = 0$$
$$2y + z = 0$$
$$11z = 0$$

Now,

$$z = 0$$
$$2y + 0 = 0$$
$$\Rightarrow y = 0$$
$$x = 0$$

Hence the system has unique solution,

$$\therefore x = 0, \; y = 0, \; z = 0 \quad \text{Ans.}$$

Eigen Vector and Eigen Values

Definition:

An Eigen Vector is a non-zero which satisfies the equation, $A\vec{V} = \lambda\vec{V}$ or, $A\vec{x} = \lambda\vec{x}$

where, A is the given square matrix

λ is the scalar value which is called the Eigen value of the given matrix.

$\vec{V}$ or $\vec{x}$ is called the Eigen vector of the given matrix.

Example: 23

Determine which of the following vectors are Eigen vectors for $A = \begin{bmatrix} 1 & 2 \\ -4 & 7 \end{bmatrix}$, showing your analysis procedure graphically.

(a) $\begin{bmatrix} 1 \\ -1 \end{bmatrix}$ (b) $\begin{bmatrix} 2 \\ 3 \end{bmatrix}$ (c) $\begin{bmatrix} 1 \\ 2 \end{bmatrix}$

(d) $\begin{bmatrix} 2 \\ 2 \end{bmatrix}$ (e) $\begin{bmatrix} 2 \\ 4 \end{bmatrix}$ (f) $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$

Solution

We ~~hav~~ know the relation between Eigen-vector and Eigen values of a matrix, $A\vec{x} = \lambda \bar{x}$

where,

A is given matrix.

λ is Eigen values of the given matrix

$\vec{x}$ is the corresponding Eigen vectors of matrix.

(a) Mathematically test,

Given, $A = \begin{pmatrix} 1 & 2 \\ -4 & 7 \end{pmatrix}$

Let, $x = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$

$$Ax = \begin{pmatrix} 1 & 2 \\ -4 & 7 \end{pmatrix} \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

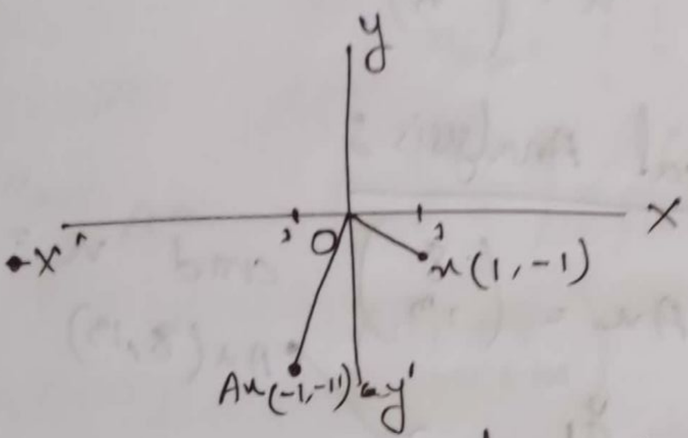
$$= \begin{pmatrix} 1\times1 - 2\times1 \\ -4\times1 - 7\times1 \end{pmatrix} = \begin{pmatrix} -1 \\ -11 \end{pmatrix} = -1 \begin{pmatrix} 1 \\ 11 \end{pmatrix}$$

As $A\vec{x} = \lambda\vec{x}$ is not satisfied. So, the given vector, $x = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$ is not Eigen vector

Ans —

(a) Graphical Analysis :

Here, $Ax = \begin{pmatrix} -1 \\ -11 \end{pmatrix}$ and $x = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$



As the product vector Ax is not in the same direction as x so, x is not Eigan vector of A.

(b) Mathematically test :

Given, $A = \begin{pmatrix} 1 & 2 \\ -4 & 7 \end{pmatrix}$

Let, $x = \begin{pmatrix} 2 \\ 3 \end{pmatrix}$

Now, $Ax = \begin{pmatrix} 1 & 2 \\ -4 & 7 \end{pmatrix} \times \begin{pmatrix} 2 \\ 3 \end{pmatrix}$

$= \begin{pmatrix} 2+6 \\ -8+21 \end{pmatrix} = \begin{pmatrix} 8 \\ 13 \end{pmatrix}$

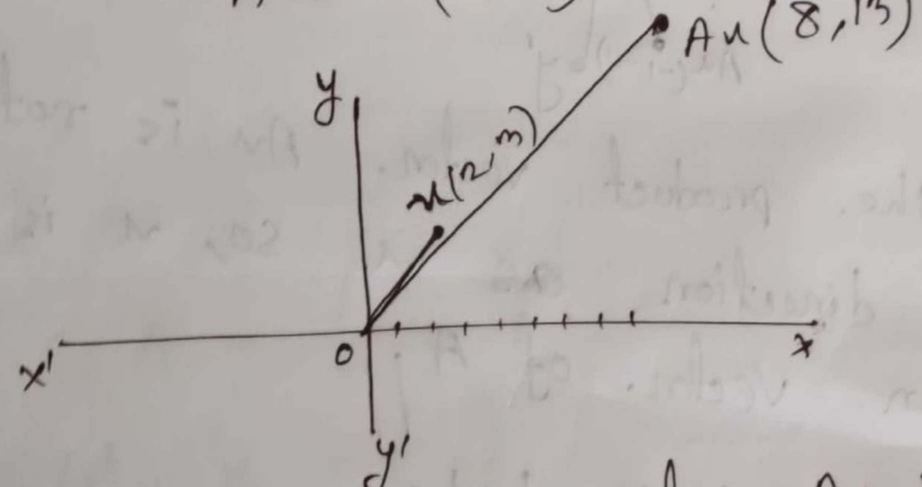
As, $A\vec{x} = \lambda\vec{x}$ is not satisfied. So,

the given vector,

$x = \begin{pmatrix} 2 \\ 3 \end{pmatrix}$ is not Eigen vector

(b) Graphical Analysis :

Here, $Ax = \begin{pmatrix} 8 \\ 13 \end{pmatrix}$ and $x = \begin{pmatrix} 2 \\ 3 \end{pmatrix}$



As, the product vector Ax is not in the same direction as vector x.

So, the given vector is not Eigen vector.

Example 24:

Find the Eigen values and Eigen vectors of matrix $A = \begin{bmatrix} 1 & 2 \\ 3 & 2 \end{bmatrix}$

Solution:

Here given, $A = \begin{bmatrix} 1 & 2 \\ 3 & 2 \end{bmatrix}$

$$A - \lambda I = \begin{bmatrix} 1 & 2 \\ 3 & 2 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 2 \\ 3 & 2 \end{bmatrix} - \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix}$$

$$= \begin{bmatrix} 1-\lambda & 2 \\ 3 & 2-\lambda \end{bmatrix}$$

So the characteristice eavation,

$$\therefore |A - \lambda I| = \begin{vmatrix} 1-\lambda & 2 \\ 3 & 2-\lambda \end{vmatrix}$$

$$= (1-\lambda) \times (2-\lambda) - 6$$

$= 2 - \lambda - 2\lambda + \lambda^2 - 6$

$= \lambda^2 - 3\lambda - 4$

$= \cancel{(\lambda^2 (\; (\lambda^2 - 4\lambda + \lambda - 4)}$

$|A - \lambda I| = 0$

$\Rightarrow (\lambda^2 - 3\lambda - 4) = 0$

$\Rightarrow (\lambda - 4)(\lambda + 1) = 0$

$\therefore \lambda = 4$ and $\lambda = -1$

Let a non-zero vector $v = \begin{bmatrix} x \\ y \end{bmatrix}$

We have, $Av = \lambda v$

$\begin{bmatrix} 1 & 2 \\ 3 & 2 \end{bmatrix} \times \begin{bmatrix} x \\ y \end{bmatrix} = \lambda \begin{bmatrix} x \\ y \end{bmatrix}$

$\Rightarrow \begin{bmatrix} 1 & 2 \\ 3 & 2 \end{bmatrix} \times \begin{bmatrix} x \\ y \end{bmatrix} = -1 \begin{bmatrix} x \\ y \end{bmatrix}$ [when $\lambda =$

$$\Rightarrow \begin{bmatrix} x+2y \\ 3x+2y \end{bmatrix} = -1 \begin{bmatrix} x \\ y \end{bmatrix}$$

Then we obtain the system of equations,

$$x + 2y = -x$$
$$3x + 2y = -y$$

Now, $x + 2y = -x$

$\Rightarrow 2x + 2y = 0$ [Dividing by 2]

$\Rightarrow x + y = 0$

And, $3x + 2y = -y$

$\Rightarrow 3x + 3y = 0$ [Dividing by 3]

$\Rightarrow x + y = 0$

we get,

$$x+y=0$$
$$x+y=0$$

Since the 1st and 2nd equations are identical. We can disregard one of them

$$x + y = 0$$

In echelon form, there are only one equation in two variable, then the system has a nonzero solution in particular (2-1) = 1 free variable, which is y.

Let, $y = 2$

$\therefore x + y = 0$

$\Rightarrow x = -2$

$v = \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} -2 \\ 2 \end{bmatrix}$ is a non-zero Eigen vector belonging to the Eigen value $\lambda = -1$

Again, for $\lambda = 4$

$Av = \lambda v$

$\Rightarrow \begin{bmatrix} 1 & 2 \\ 3 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \lambda \begin{bmatrix} x \\ y \end{bmatrix}$

$\Rightarrow \begin{bmatrix} 1 & 2 \\ 3 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = 4 \begin{bmatrix} x \\ y \end{bmatrix}$

$\Rightarrow \begin{bmatrix} x + 2y \\ 3x + 2y \end{bmatrix} = \begin{bmatrix} 4x \\ 4y \end{bmatrix}$

Then we obtain the system of equations

$x + 2y = 4x$

$\Rightarrow -3x + 2y = 0$

$\Rightarrow 3x - 2y = 0$

and $3x + 2y = 4y$

$\Rightarrow 3x - 2y = 0$

We get,

$3x - 2y = 0$

$3x - 2y = 0$

Since the 1st and 2nd equations are identical.

we can ~~dis~~ ~~delt~~ delete one of them.

So, $3x - 2y = 0$

In echelon form, there are only one equation ~~and two~~ in two unknowns, then the system has a non-zero solution and in particular $(2-1) = 1$ free variable, which is y.

Let, $y = 3$

$\therefore 3x - 2y = 0$

$\Rightarrow 3x - 2 \times 3 = 0$

$\Rightarrow 3x = 6$

$\therefore x = 2$

$\therefore v = \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$ is a non-zero eigen vector belonging to the eigen value $\lambda = 4$

$\therefore -1, 4$ are the eigen values of the matrix A and their respective eigen vectors are $\begin{bmatrix} -2 \\ 2 \end{bmatrix}$ and $\begin{bmatrix} 2 \\ 3 \end{bmatrix}$.

(most important)

Example - 28

Definition of Diagonal matrix or $\boxed{\text{only matrix}}$ and Diagonalize the matrix $\begin{pmatrix} 4 & 1 \\ 1 & 4 \end{pmatrix}$

Solution:

Definition of Diagonal matrix:

A square matrix in which all elements are zero, Expect those in the leading diagonal is known as a diagonal matrix

$$A = \begin{bmatrix} a & 0 & 0 \\ 0 & b & 0 \\ 0 & 0 & c \end{bmatrix} \quad \cancel{\text{where}} \; a \neq b \neq c$$

Formula :

$$D = P^{-1} A P$$

whene D is diagonal matrix

A is given matrix

P is the matrix of Eigen vector

P^{-1} is the inverse matrix of P

Formula : If $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ then,

$$A^{-1} = \frac{1}{ad-bc} \times \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}, \text{~~then~~}$$

<u>Solution</u>

Let, $A = \begin{bmatrix} 4 & 1 \\ 1 & 4 \end{bmatrix}$

$$\therefore A - \lambda I = \begin{bmatrix} 4 & 1 \\ 1 & 4 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 4 & 1 \\ 1 & 4 \end{bmatrix} - \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix}$$

$$= \begin{bmatrix} 4-\lambda & 1 \\ 1 & 4-\lambda \end{bmatrix}$$

So the characteristic equation,

$$|A - \lambda I| = \begin{vmatrix} 4-\lambda & 1 \\ 1 & 4-\lambda \end{vmatrix}$$

$$= (4-\lambda) \times (4-\lambda) - 1$$

$$= 16 - 4\lambda - 4\lambda + \lambda^2 - 1$$

$$= \lambda^2 - 8\lambda + 15$$

Now, $|A - \lambda I| = 0$

$\Rightarrow \lambda^2 - 8\lambda + 15 = 0$

$\Rightarrow (\lambda - 3)(\lambda - 5) = 0$

$\therefore \lambda = 3, 5.$

Let a non-zero vector $v_1 = \begin{bmatrix} x \\ y \end{bmatrix}$

we have,

$$Av_1 = \lambda v_1$$

$$\begin{bmatrix} 4 & 1 \\ 1 & 4 \end{bmatrix} \times \begin{bmatrix} x \\ y \end{bmatrix} = 3\begin{bmatrix} x \\ y \end{bmatrix} \quad [\text{when } \lambda = 3]$$

$$\Rightarrow \begin{bmatrix} 4x + y \\ x + 4y \end{bmatrix} = \begin{bmatrix} 3x \\ 3y \end{bmatrix}$$

Then we obtain the system of equation.

$$4x+y = 3x$$
$$\Rightarrow 4x+y-3x = 0$$
$$\Rightarrow x+y = 0$$

and $x + 4y = 3y$

$$\Rightarrow x + y = 0$$

we get

$$x+y=0$$
$$x+y=0$$

Since the 1st and 2nd equations are identical we can delete one of them.

$$x + y = 0$$

In echelon form, there are only one equation in two unknowns, then the system has a non-zero solution and in particular $(2-1)=1$ free variable, which is y.

Let, $y=1$, $\therefore x=-1$

$\therefore v_1 = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$ → Let, $v_2 = \begin{bmatrix} x \\ y \end{bmatrix}$

We have, $Av_2 = \lambda v_2$

$\Rightarrow \begin{bmatrix} 4 & 1 \\ 1 & 4 \end{bmatrix} \times \begin{bmatrix} x \\ y \end{bmatrix} = 5 \begin{bmatrix} x \\ y \end{bmatrix}$ [when, $\lambda = 5$]

$\Rightarrow \begin{bmatrix} 4x+y \\ x+4y \end{bmatrix} = \begin{bmatrix} 5x \\ 5y \end{bmatrix}$

Then we obtain the system of equations,

$4x+y=5x$ and $x+4y=5y$

$\Rightarrow 4x-5x+y=0$ $\Rightarrow x+4y-5y=0$

$\Rightarrow -x+y=0$ $\Rightarrow x-y=0$

$\Rightarrow x-y=0$

$\therefore$ we get,

$x - y = 0$
$x - y = 0$

Since the 1^{st} and 2nd equations are identical. We cand delete one of them

$\therefore x - y = 0$

In echelon form, there are only one equation in two unknowns, then the system has a non-zero solution and in particular $(2-1) = 1$ free variable, which is y.

Let $y = 1$, $\therefore x = 1$

$$\therefore P = \begin{bmatrix} v_1, & v_2 \end{bmatrix}$$

$$= \begin{bmatrix} -1 & 1 \\ 1 & 1 \end{bmatrix}$$

$$= -1 - 1$$
$$= -2$$

Therefore,

$$P^{-1} = \frac{1}{-1-1}\begin{bmatrix} 1 & -1 \\ -1 & -1 \end{bmatrix}$$

$$= \frac{1}{-2}\begin{bmatrix} 1 & -1 \\ -1 & -1 \end{bmatrix}$$

$$= \begin{bmatrix} -1/2 & 1/2 \\ 1/2 & 1/2 \end{bmatrix}$$

Hence the required diagonal matrix,

$$D = P^{-1}AP$$

$$= \begin{bmatrix} -\frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} \end{bmatrix} \times \begin{bmatrix} 4 & 1 \\ 1 & 4 \end{bmatrix} \times \begin{bmatrix} -1 & 1 \\ 1 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} -1/2 & \frac{1}{2} \\ 1/2 & 1/2 \end{bmatrix} \times \begin{bmatrix} -4+1 & 4+1 \\ -1+4 & 1+4 \end{bmatrix}$$

$$= \begin{bmatrix} -1/2 & 1/2 \\ 1/2 & 1/2 \end{bmatrix} \times \begin{bmatrix} -3 & 5 \\ 3 & 5 \end{bmatrix}$$

$$= \begin{bmatrix} 3/2 + 3/2 & -5/2 + 5/2 \\ -3/2 + \frac{3}{2} & \frac{5}{2} + \frac{5}{2} \end{bmatrix}$$

$$= \begin{bmatrix} 3 & 0 \\ 0 & 5 \end{bmatrix} \text{ Ans-}$$

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Home task

Finding Inverse using Equations:

Find A^{-1} where $A = \begin{bmatrix} 1 & 2 \\ 3 & 1 \end{bmatrix}$

Solution:

Given, $A = \begin{bmatrix} 1 & 2 \\ 3 & 1 \end{bmatrix}$,

Let, $X = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$

and $B = \begin{bmatrix} d_1 \\ d_2 \end{bmatrix}$

$\therefore AX = B$

$\begin{bmatrix} 1 & 2 \\ 3 & 1 \end{bmatrix} \times \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} d_1 \\ d_2 \end{bmatrix}$

$\Rightarrow \begin{bmatrix} x_1 + 2x_2 \\ 3x_1 + x_2 \end{bmatrix} = \begin{bmatrix} d_1 \\ d_2 \end{bmatrix}$

The matrix can be written as

$x_1 + 2x_2 = d_1$ ——— (i)

$3x_1 + x_2 = d_2$ ——— (ii)

$\therefore$ (ii) $\times$ 2 $-$ (i) $\Rightarrow$

$$2 \times 3x_1 + \cancel{2x_2} = 2d_2$$
$$-\quad x_1 \underset{(-)}{+} \cancel{2x_2} = \underset{(-)}{d_1}$$

$$\Rightarrow 5x_1 = 2d_2 - d_1$$

$$\Rightarrow x_1 = -\frac{1}{5}d_1 + \frac{2}{5}d_2 \quad \text{—— (iii)}$$

$\therefore$ (i) $\times$ 3 $-$ (ii) $\Rightarrow$

$$\cancel{3x_1} + 6x_2 = 3d_1$$
$$-\ \cancel{3x_1} - x_2 = -d_2$$

$$5x_2 = 3d_1 - d_2$$

$$\Rightarrow x_2 = \frac{3}{5}d_1 - \frac{1}{5}d_2 \quad \text{—— (iv)}$$

from (iii) and (iv) we get,

$$\therefore A^{-1} = \begin{bmatrix} -\frac{1}{5} & \frac{2}{5} \\ \frac{3}{5} & -\frac{1}{5} \end{bmatrix}$$

CamScanner

• Define Augmented matrix with example

[4] **Augmented Matrix :**

The augmented matrix for $AX = b$ is partitioned matrix $[A|b]$.

Example 30:

$$x_1 + x_2 - 2x_3 = -3$$

$$2x_1 + 5x_2 + 3x_3 = 11$$

$$-x_1 + 3x_2 + x_3 = 5$$

can be written as the matrix equation

$$\begin{bmatrix} 1 & 1 & -2 \\ 2 & 5 & 3 \\ -1 & 3 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} -3 \\ 11 \\ 5 \end{bmatrix}$$

which has as its augmented matrix ?

$$[A|b] = \left[\begin{bmatrix} 1 & 1 & -2 \\ 2 & 5 & 3 \\ -1 & 3 & 1 \end{bmatrix} \begin{matrix} -3 \\ 11 \\ 5 \end{matrix} \right]$$

st

Linear Independent and Linear Dependent

Definition: A ~~collection~~ of vectors is either

~~the~~ ~~fall~~ ~~linearly~~

~~The~~

Linear independent:

The vectors $\{v_1, v_2, v_3, v_4 \cdots, v_n\}$ are linearly independent if the equation involving linear combinations

$a_1v_1 + a_2v_2 + a_3v_3 + a_4v_4 + \cdots + a_nv_n = 0$ is true only when the scalars $a_i = (a_1, a_2, a_3 \cdots, a_n)$ are all equal to zero.

Linearly dependent:

The vectors are linearly dependent if the equation has a solution when at least one of the scalars is not zero.

Example - 33:

Determine whether the following vectors in R^2 are linearly dependent or linearly independent

$x_1 = \begin{bmatrix} -1 \\ 3 \end{bmatrix}$, $x_2 = \begin{bmatrix} 5 \\ 6 \end{bmatrix}$, $x_3 = \begin{bmatrix} 1 \\ 4 \end{bmatrix}$

Solution:

Suppose we have a linear combination of the vector equal to 0

$$\therefore a_1 x_1 + a_2 x_2 + a_3 x_3 = 0$$

$$\Rightarrow a_1 \begin{bmatrix} -1 \\ 3 \end{bmatrix} + a_2 \begin{bmatrix} 5 \\ 6 \end{bmatrix} + a_3 \begin{bmatrix} 1 \\ 4 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} -a_1 \\ 3a_1 \end{bmatrix} + \begin{bmatrix} 5a_2 \\ 6a_2 \end{bmatrix} + \begin{bmatrix} a_3 \\ 4a_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} -a_1 + 5a_2 + a_3 \\ 3a_1 + 6a_2 + 4a_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\therefore -a_1 + 5a_2 + a_3 = 0 \text{ ——— } L_1$$

$$3a_1 + 6a_2 + 4a_3 = 0 \text{ ——— } L_2$$

To reduce it to a simpler system for each $i > 1$ applies operation :

We know,

$$L_i = -a_{i1}L_1 + a_{11}L_i$$

Let, $i = 2$,

$$L_2 = -a_{21}L_1 + a_{11}L_2$$

$$= -3(-a_1 + 5a_2 + a_3) + (-1)(3a_1 + 6a_2 + 4a_3)$$

$$= 3a_1 \cancel{+} 15a_2 - 3a_3 - 3a_1 - 6a_2 - 4a_3$$

$$= -21a_2 - 7a_3$$

Thus we obtain the following new system

$$-a_1 + 5a_2 + a_3 = 0$$

$$-21a_2 - 7a_3 = 0$$

In echelon form, there are only two equation in three unknown; then the system has a non-zero solution and in particular $(3-2) = 1$ free variable, which is a_3.

Since, a_3 is free variable, we can choose it to be anything. In particular, we can choose it to be non-zero

Therefore, the vectors are linearly dependent.

Example 34:

Determine whether the following vectors in R^3 are linearly dependent or independent.

$$x_1 = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, \quad x_2 = \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix}, \quad x_3 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$$

Solution

Suppose we have a linear combination of the vectors equal to 0.

$$a_1 x_1 + a_2 x_2 + a_3 x_3 = 0$$

$$\Rightarrow a_1 \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} + a_2 \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix} + a_3 \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} a_1 \\ 2a_1 \\ 3a_1 \end{bmatrix} + \begin{bmatrix} -2a_2 \\ a_2 \\ 0a_2 \end{bmatrix} + \begin{bmatrix} a_3 \\ 0a_3 \\ a_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} a_1 - 2a_2 + a_3 \\ 2a_1 + a_2 + 0a_3 \\ 3a_1 + 0a_2 + a_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$a_1 - 2a_2 + a_3 = 0 \longrightarrow L_1$$

$$2a_1 + a_2 + 0a_3 = 0 \longrightarrow L_2$$

$$3a_1 + 0a_2 + a_3 = 0 \longrightarrow L_3$$

$$L_i = -a_{i1}L_1 + a_{11}L_i \quad \text{—— (1)}$$

Let $i = 2$,

$$L_2 = -a_{21}L_1 + a_{11}L_2$$
$$= -2(a_1 - 2a_2 + a_3) + 1 \times (2a_1 + a_2)$$
$$= -2a_1 + 4a_2 - 2a_3 + 2a_1 + a_2$$
$$= 5a_2 - 2a_3$$

~~Let $i = 3$,~~ ~~Now equation will be,~~

$\cancel{3a_1 + 0a_2 + a_3 = 0}$

$5a_2 \cancel{- 2a_3 = 0} \longrightarrow L_1$

Let, $i = 3$ in eqn (1),

$$L_3 = -a_{31}L_1 + a_{11}L_3$$
$$= -3(a_1 - 2a_2 + a_3) + 1 \times (3a_1 + a_3)$$
$$= -3a_1 + 6a_2 - 3a_3 + 3a_1 + a_3$$

$= 6a_2 - 2a_3$

Thus we obtain the following new System,

$$a_1 - 2a_2 + a_3 = 0$$
$$5a_2 - 2a_3 = 0 \text{ —— } L_1$$
$$6a_2 - 2a_3 = 0 \text{ —— } L_2$$

Now putting $i = 2$ in $eq^n (1) \Rightarrow$

$$L_2 = -a_{21}L_1 + a_{11}L_2$$
$$= -6(5a_2 - 2a_3) + 5\times(6a_2 - 2a_2)$$
$$= -30a_2 + 12a_3 + 30a_2 - 10a_2$$
$$= 2a_3$$

Thus we obtain following new System,

$$a_1 - 2a_2 + a_3 = 0$$
$$5a_2 - 2a_3 = 0$$
$$2a_3 = 0$$

So, $a_3 = 0$; $a_2 = 0$, $a_1 = 0$

Therefore, the vectors are linearly independent